Ingredients of Ellipsis: E-extension and the Uniformity of Silence*

Ido Benbaji-Elhadad (ibenbaji@mit.edu)
David Pesetsky (pesetsk@mit.edu)

Massachusetts Institute of Technology

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Abstract

This paper presents a new general proposal concerning the syntax of ellipsis. We argue that phrasal ellipsis is always preceded by, and depends on, some independent process that silences the head of the constituent targeted for deletion. Specifically, we propose that all syntactic processes that silence an otherwise overt expression do so by applying an **E-feature** to that expression (**Uniformity of Silence**). With this as background, we argue that (at least some) phenomena traditionally studied under the special rubric of "ellipsis" involve nothing more than a process of **E-extension**, which extends the E-feature independently assigned to a head H° to a slightly higher node — the minimal node that contains both H° and its selected complement.

We propose that the phenomenon known as **Sluicing** is simply the result of the E-feature assigned to a complementizer in a "Doubly-Filled Comp Effect" configuration, extended to the minimal C′ projection. Furthermore, the family of phenomena usually studied under the rubric of **VP-Ellipsis**, including English Auxiliary Verb Phrase Ellipsis, is argued to result from extension of the E-feature that silences the trace of verb-raising to the minimal V′ or AUX′ containing that verb. Crucially, silencing in both cases does not target a maximal projection, but rather a projection minimally larger than the silenced head, allowing us to unify the analysis of Adjunct Exclusion effects in Verb-stranding VPE configurations (Landau, 2018, 2020) with the behavior of discourse particles in Sluiced CPs (Marušič et al., 2015, 2019). In the verbal domain in particular, our proposal predicts a correlation between the possibility of ellipsis and the availability of head movement — a correlation which is attested in English AuxP ellipsis. It also provides a new perspective on the distribution of English expletive *do* and on the syntax of voice mismatches (adapting proposals by Merchant 2013).

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1 Goals

This paper presents and defends a new hypothesis about the nature of phenomena usually described as "ellipsis". In particular, we propose a common syntactic source for situations in which an entire phrase is silenced that is independent of the identity of that phrase.

Our proposal begins with the conjecture that all syntactic processes that silence an otherwise overt expression do the same thing: apply a feature "**E**" to a position in syntactic structure, whose effect is to silence its contents. We call this the *Uniformity of Silence Conjecture*.

(1) **Effect of E-feature**
The feature E associated with a node α silences every element reflexively dominated by α.

(2) **Uniformity of Silence Conjecture**
A phrase or lexical item is silent if and only if it bears the feature E.

It is with malice aforethought that we have named the relevant silencing feature "E", the same name as the feature posited by Merchant (2001), Aelbrecht (2010) and many others as an essential component of the analysis of ellipsis. In the spirit (but not the letter) of Landau (2020a), however, we propose that E silences the element that bears it, rather than just its complement or some other near-by element, as Merchant and most others have proposed.

With the Uniformity of Silence Conjecture as background, we narrow our focus to silent syntactic *heads* and argue that at least some phenomena traditionally studied under the special rubric of *ellipsis* involve nothing more than extending an E-feature on a syntactic head to include that head's complement; a process that we call "**E-extension**".

We will discuss two distinct situations in which syntactic heads are silent, and argue that in each case, the optional process of E-extension yields a well-known instance of ellipsis. Our first example concerns the Doubly-Filled Comp effect, which we propose arises from a featural co-occurrence restriction regulating the distribution of complementizers with and without E. We will argue that the kind of ellipsis known as *sluicing* is the simple result of E-extension from the complementizer to the minimal projection that includes the subcategorized complement of C as well as C itself, namely the node conventionally called C′. Our second example concerns the silencing of traces of verbal head movement, which we propose involves the assignment of E to the trace position.[1] Here, we will argue that extension of E to the minimal projection that includes both the original position of the moved verb and its complement, i.e. V′, yields the full range of constructions usually identified as *VP ellipsis* — which we propose are actually instances of generalized verb stranding verbal projection ellipsis.

Both instances of E-extension crucially target a phrase no larger and no smaller than the smallest phrase containing the head and its complement. This yields a silent phrase that crucially excludes elements higher than the single-bar projection of the E-bearing head, making correct predictions in both the domain of sluicing and so-called VP-ellipsis. In the domain of sluicing, the proposal predicts a distinction between elements that must be included in the silenced material (including heads moved to C as well as the complementizer itself and its complement) and dependents of the C-system that are not silenced (specifiers of CP and high discourse particles). In the domain of phenomena traditionally analyzed as VP ellipsis, we will argue that the same proposal derives the effect dubbed *Adjunct Exclusion* by Landau (2018), and extends to a wider range of phenomena as well. In particular, any element outside the single-bar projection of the E-bearing trace of head movement is predicted to remain unsilenced by the operation of E-extension. We show that this property of our proposal is exemplified not only by the exclusion of adjuncts, but also by the exclusion of higher negation and focus-sensitive operators from domains silenced as a consequence of verb movement and E-extension — with examples from English and Hebrew.

Crucially, the reasons why a head might be assigned an E-feature are heterogeneous (the Doubly-Filled Comp effect, silencing of traces), but the consequences of the process of E-extension are uniform: the broader silencing traditionally described as ellipsis, with a uniform set of properties such as Adjunct Exclusion. Ellipsis, we thus argue, is not a thing unto itself, but parasitic on the silencing of heads.

1. The conjecture that the silencing of a trace of movement and the silencing of an elided phrase arise from the same instruction from the syntax to phonetics/phonology was first advanced (we believe) by Chomsky (1995, 186), and has been explored in a different fashion from the present paper in a number of publications by Saab (2010; 2022b; among others). We are grateful to Idan Landau for pointing us to these predecessors.

The uniform account of ellipsis phenomena that we propose here can now be used to diagnose the presence of silenced heads in constructions whose analysis has previously been in some doubt. In the final sections of this paper, we defend the idea that *all* instances of apparent VP-ellipsis, including apparent AuxP-ellipsis in English, result from E-extension — by showing that the possibility of ellipsis within the English auxiliary system correlates with the availability (and unavailability) of short movement of the verbs within that system. This observation in turn will guide us towards a slightly novel structure for the English auxiliary system as a whole (developing earlier work by Harwood 2014). Our analysis of the English auxiliary system in turn will permit a significantly more unified account of the distribution of expletive *do*, and sheds light on the analysis of voice and voice mismatches in ellipsis, as discussed by Merchant (2013) among others.

Our discussion throughout this paper presupposes a separation between the laws of syntax and phonological interpretation responsible for the existence and distribution of ellipsis configurations and the laws that govern the interpretation of silenced material — and focuses almost entirely on the first of these topics. There is of course a large and constantly growing literature devoted to the semantic properties of ellipsis configurations, exploring parallelism requirements, interactions with scope and anaphoric relations, and much more. We will assume at a minimum that silenced material must have an antecedent that supplies its interpretation — and that *syntactic identity* of some sort is required. This will be a crucial assumption when we turn our attention to Adjunct Exclusion in verbal ellipsis configurations, and will be an essential premise of our brief discussion of voice mismatches in section 9 at the end of this paper. At the same time, we expect that our overall approach to the syntax of ellipsis phenomena will prove compatible with a variety of proposals concerning the precise nature of their antecedency and parallelism requirements, but leave a proper investigation of those connections for future work.

Part I: Sluicing

2 Doubly-Filled Comp phenomena

In Standard English and many other languages, it has been observed that a complementizer must be unpronounced in the presence of certain kinds of specifiers. This observation was first discussed in detail by Keyser (1975, 27 ex. 66) and taken up by Chomsky & Lasnik (1977, 446 ex. (53)), who used the now rather opaque term **Doubly-Filled Comp** (henceforth **DFC**) to describe the configuration in which this effect is observed.[2]

(3) **DFC effect in Standard English questions: complementizer must be silenced**
   a. *Sue asked [**who that** Mary met ___].
   b. Sue asked [**who** ~~that~~ Mary met ___].

In the sections that follow, we will argue that the DFC effect is the result of a feature co-occurrence rule that regulates the use of silent and overt variants of the complementizer — complementizers with and without an E-feature — in a particular environment. We begin with a characterization of that environment and a justification of the logic of our proposed account for DFC effects — including the proper treatment of languages that permit some variation in complementizer choice. Though some aspects of our characterization of the DFC effect are new, explaining the phenomenon is not a primary goal of this paper. We will not discuss why the DFC effect might exist in the first place, and we will not derive its properties from deeper principles. But once we have an explicit description of the phenomenon in hand, we will argue that it brings us within a hair's breadth of explaining the existence of sluicing, along with many of its puzzling properties. That will be the topic of the section that follows this one.

Specifier-induced DFC effects. In some languages such as Modern English and formal varieties of European French, the DFC effect involves the obligatory use of a phonologically null complementizer in a specific syntactic environment. In other languages such as some varieties of Dutch, Middle English, and other varieties of French, the same

2. The term "Doubly Filled Comp" coined by Chomsky & Lasnik (1977, 461) made sense in the context of their analysis of clause structure, which did not posit a normal headed X-bar structure for clauses, but instead countenanced a non-projecting comp node stipulated to immediately dominate two positions, one reserved for a *wh*-phrase and one for the complementizer. As so often, the term has outlived the context in which it made sense, adapted in the obvious fashion to the better-supported structure for the clause in which C is the head of the clause, taking the rest of the clause as its sister, with *wh*-phrases and similar elements functioning as specifiers.

environment merely licenses the optional use of a phonologically null complementizer that is otherwise obligatorily overt. We focus first on languages of the first type, then turn to the second.

The most commonly discussed syntactic environment that triggers the obligatory use of a silent complementizer in English is characterized by the presence of a *wh*-specifier, as in (3). Previous work on the DFC effect has generally proposed that the specifier must be phonologically *overt* to trigger a DFC effect. Keyser (1975), for example, viewed the relative clause paradigm in (4) as evidence for the relevance of overtness to the DFC effect — since the use of an overt complementizer, banned as an instance of the DFC effect in the presence of an overt relative operator in (4a), is allowed when the relative operator is unpronounced in (4b).[3]

(4) **DFC effect in Standard English relative clauses: possible relevance of overtness of specifier**
  a. the person who (*that) we saw
  b. the person $ø_{rel}$ (✓that) we saw

Similarly, though there is ample evidence that Ā-movement across a clause boundary must make a stop at its edge, forming an intermediate specifier of CP, this intermediate specifier does not trigger the obligatory silencing of a complementizer — a fact that we might be tempted to attribute to its status as an unpronounced trace:

(5) **No DFC effect induced by silent intermediate positions**

Mary asked [who (*that) John believed [— that we might invite —]
DFC effect *no DFC effect*

At the same time, one well-known argument for the existence of such intermediate specifier positions in the first place is the fact that some of its subconstituents may sometimes be pronounced overtly — arguably as an instance of "distributed deletion" (Fanselow & Ćavar 2002). When this happens, the intermediate specifier still fails to trigger a DFC effect. The complementizer may remain overt even though its specifier contains overt material. One example is provided by the West Ulster dialect of Irish English, where an overt pluralizing morpheme *all* associated with a moved *wh*-phrase can be pronounced in intermediate specifiers of CP (a discovery due to McCloskey 2000). This overt material, though contained within the specifier position of CP, triggers no DFC effect in either finite or nonfinite clauses, as shown in in (6a-b). A similar point can be made with quantity modifier stranding in Standard English, seen in (6c); and in Russian with prepositions that can also be used postpositionally (Podobryaev 2009), as discussed by Davis (2020) and illustrated in (6d).[4]

(6) **DFC effect limited to "criterial" C: no effect induced by intermediate positions with overt material**
West Ulster pluralizer stranding
  a. Who did Frank tell you [$_{CP}$ [___ **all**] **that** they were after ___ ?
  'Who.PL did Frank tell you that they were after?'
  b. Who did you arrange [$_{CP}$ [___ **all**] **for** your mother to meet ___ at the party]?
  'Who.PL did you arrange for your mother to meet at the party'
(McCloskey 2000, 61 ex. (9d), 70 ex. (41a))

3. We return to the version of (4b) in which *that* is unpronounced below.

4. Colloquial English possessum stranding appears at first sight to counterexemplify this claim, as discussed by Davis (2021, 23) — who also notes the contrast with constructions like (6a-b) (p. 33, fn. 29).

(i) Who do they think [(*that) [___ 's name] (*that) is Mary]? (Davis 2021, 20 ex. (50b))

It is possible, however, that the stranded nominal in this construction actually occupies a topic or focus position lower than C, from which the possessor is subextracted.

Polish data discussed by Wiland (2010, 342) seem to pose a similar counterexample, if analyzed as stranding of a nominal at the left edge of CP by *wh*-movement that does force the silencing of the complementizer. The examples reported, however, permit a parenthetical parse of what was intended as a matrix clause embedding a declarative *that*-clause. Replacing this material with alternatives that block a parenthetical parse degrades the entire construction — possibly removing it as an example of stranding in spec,CP in the first place (though they remain somewhat worse with an overt complementizer). We leave this as a puzzle for future research.

English quantity modifier stranding

c. Tell me [how much flour you said [$_{CP}$ [___ **to the nearest pound**] **that** the bakery wants ___]]
(Davis 2020, 4 ex. (11))

Russian postposition stranding

d. ?Komu Vasja xotel [$_{CP}$ [___ **navstreču**] **čtoby** Petja pobežal ___ ?]
who.DAT Vasja wanted towards COMP.SBJV Petja run.SBJV
'Toward whom did Vasja want Petja to run?' (Davis 2020, 10 ex. (30))

This observation suggests that a specifier that triggers a DFC effect must be a final rather than intermediate position — the property that Rizzi (2006) called *criteriality*. Whether status as a criterial position is a sufficient or merely necessary condition for a spec,CP to trigger a DFC effect depends on the analysis of examples like (4b). If the specifier of an English *that*-relative is the final landing site for an phonologically null relative operator, as portrayed in that example, then overtness as well as criteriality must figure in any statement of the environment that yields the DFC effect. If, on the other hand, the specifier of the relative clause is merely an intermediate landing site for the raising of the head NP (so that the position labeled "$\emptyset_{rel}$" is actually a trace of the NP *person*), then overtness of the specifier can be considered irrelevant to the DFC effect, and all that matters is criteriality. For the purposes of this paper, we will assume the latter is the case — that a specifier's overtness is indeed irrelevant to the DFC effect, contrary to the traditional view — our proposals are also compatible with the more traditional alternative. For specific reasons that will become clearer below, and in keeping with the Uniformity of Silence Conjecture, we represent the phonological specification of a silent element as the feature E:

(7) **DFC effect, version 1 (English)**
A complementizer bears E (and is therefore unpronounced) obligatorily if its specifier is criterial.

The statement in (7) leaves open the analysis of English embedded finite declarative clauses that seem to show neither an overt complementizer nor a specifier of CP, such as the embedded clause of sentences like *Sue claimed the student would arrive*. If the embedded clause in such constructions is a full CP with a silent complementizer and no specifier, then we must hunt for a reason other than (7) why a complementizer can be silent. In fact, however, Pesetsky & Torrego (2001, 373 ff.) and Pesetsky (2019) have argued that the silence of the complementizer in such examples actually is a DFC effect.[5] Specifically, they proposed that the subject undergoes short-distance movement to form a specifier of CP in response to an optional φ-probe on C — and that this specifier of CP position is criterial, i.e. the subject's final position. The subject thus occupies the same position as the *wh*-phrase in (3). As an argument in favor of the raising of the subject in such constructions, they noted (building on Doherty 1997 and Grimshaw 1997) that sentence-level adverbs that can intervene between the subject and overt complementizer *that* must follow the subject and may not precede it when the complementizer is unpronounced:

(8) **Adverbials may not precede subject in complementizerless declarative clauses (English)**
a. Sue thought [*(that) soon the student would arrive].
cf. *Sue thought* [*the student soon would arrive*].
b. She insisted [*(that) most of the time they accepted this solution].
(Grimshaw 1997, 411, ex. (43a))
cf. *She insisted* [*they most of the time accepted this solution*].
c. Mary is claiming [*(that) for all intents and purposes John is the mayor of the city].
(Pesetsky & Torrego 2001, 375, ex. (37b))
cf. *Mary is claiming* [*John for all intents and purposes is the mayor of the city*].

5. This proposal converges nicely with the recent experimental finding by Momma et al. (2024) that the silent complementizer in a *wh*-relative (*the person who C Mary met*) structurally primes the silent complementizer in a clausal complement to V (extending earlier priming results that showed that silent complementizers in clausal complements interact with other silent complementizers, but not their overt counterparts). The proposal discussed here treats both phenomena as instances of the DFC effect, so that both involve the same silent element.

This pattern can be explained if the silence of the complementizer entails that the subject has raised to Spec,CP, a position higher than the highest position that the adverbial can occupy within the clause:[6]

(9) **Analysis of silent *that* in embedded declaratives**

She claimed [the student C [soon — would arrive]]

If this approach is correct, we can (at least speculatively) strengthen (7) to (10):

(10) **DFC effect, version 2 (English)**
A complementizer bears E obligatorily if its specifier is criterial, otherwise it does not bear E (and thus is overt).

A language that obeys (10) but lacks a φ-probe on C capable of raising the subject to Spec,CP will be a language that shows the paradigm in (3) in comparable interrogative and relative clauses, but lacks the option of suppressing pronunciation of the complementizer in embedded declarative clauses. Formal European French is such a language, requiring a silent complementizer in an interrogative or relative clause introduced by a *wh*-phrase, but requiring an overt complementizer otherwise (in finite clauses, at least): *Marie pense *(que) l'étudiante est arriveé* 'Marie thinks *(that) the student has arrived'.

We have focused so far on the status of the complementizer in finite clauses, Extending the discussion to nonfinite clauses would require a more complex discussion than is possible here; as would a discussion of the silencing of complementizers (obligatory in some languages such as English) when the subject is extracted from its clause (so-called *that*-trace effects). We will leave these issues as loose ends in this paper, but see Pesetsky (2019) for discussion of factors other than the DFC effect that may be at play in infinitival raising and control constructions. We will also assume that matrix declarative clauses that are not introduced with an overt complementizer are smaller than CP in the first place, rendering them irrelevant to the issue of DFC effects.

Optional DFC effects. In the discussion so far, we have diagnosed DFC effects in languages like English by the obligatory non-overtness of the complementizer in DFC environments and by the obligatory use of the overt complementizer elsewhere (assuming that our analysis of embedded *that*-less declarative clauses is correct, albeit with the caveats noted in the preceding paragraph). The limitation of a silent complementizer to specifically DFC contexts is, however, also observable in languages where it is not obligatory. Certain dialects and informal registers of Dutch are a well-known example — where we can diagnose the use of the silent complementizer as a DFC effect thanks to the fact that it is *only* in DFC environments that the Dutch finite complementizer *dat* is permitted to bear E, rendering it silent.

As seen in (11), a Dutch complementizer such as *dat* or the specifically interrogative *of* (depending on dialect) may be overt in the presence of an overt interrogative *wh*-phrase — in contrast to Standard English, where this is not possible. Crucially, it may also be silent in this DFC environment.

(11) **Optional silent complementizer in DFC environment (Dutch dialects and informal registers)**

a. Jan vroeg [$_{CP}$ wat (of) [$_{TP}$ mijn zuster ___ gelezen heeft]].
Jan asked what COMP my sister read has
'Jan asked what my sister has read.' (Broekhuis & Corver 2019, 1216 ex. (34b))

b. Vertel niet [$_{CP}$ wie (dat) [$_{TP}$ ze geroepen hebben.]]
tell not who COMP they called have
'Don't tell me who they have called.' (Barbiers 2009, 1612 ex. (5c))

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6. As Pesetsky (2019) notes, the illicit variants of (8) would be acceptable if the adverbials themselves could occupy a very high position to the left of spec,CP. It is therefore important to note that they cannot: *Sue claimed (*soon) that Mary would arrive*, etc. Strikingly, however, Doherty (1997) pointed out that some adverbials may occur in this position, and precisely this set of adverbials may precede the subject in embedded clauses not introduced by *that*:

(i) She says [when we get home (that) things will be different].
(ii) I believe [next year (that) she'll be fine].
(iii) I suppose [ordinarily (that) you would go somewhere else].
(iv) He thinks [in some circumstances (that) things would be better]. (Doherty 1997, 203, ex. (16)-(17))

The DFC environment is a necessary condition for the silencing of the Dutch finite complementizer, however. When the Dutch complementizer does not have a criterial specifier, it is not silenced.

(12) **Complementizer obligatorily overt when lacking criterial specifier (Dutch)**
Jan zegt [$_{CP}$ *(dat) Peter ziek is]
Jan says COMP Peter ill is
'Jan says that Peter is ill.'

The logic of the phenomenon overall thus suggests that the DFC effect arises from a rule shared by multiple languages whose obligatoriness is parameterized.

(13) **DFC effect, version 3**
A complementizer bears E {obligatorily [English], optionally [Dutch]} if its specifier is criterial, otherwise it does not bear E.

Head movement to C and the logic of the DFC effect. We now consider the interaction of the DFC effect with head movement. It will be important to the proposals that follow that head movement exists, and yields a structure in which the moving head and the head it targets form a unit that itself has the syntactic status of a head (an X°) rather than a phrase, yielding the structure [$_{X^o}$ Y° X°]. This is of course the most widely accepted proposal for the output of head movement since the earliest systematic studies of the topic such as Travis (1984, 130 ff.), but has been the topic of controversy and debate nonetheless (a discussion well-surveyed by Dékány 2018). To the extent that the present paper makes a good case for its central claims but must assume this output structure for head movement, it will constitute an actual argument for the proposal. With this as background, we turn to an observation that will be central to our analysis of sluicing as grounded in the DFC effect.

The DFC effect crucially concerns only the overtness of the complementizer morpheme itself, not the overtness of C as a whole. It does not mandate silence for a head that moved to C from another position, such as a main or auxiliary verb. In an English matrix *wh*-question, for example, the complementizer must be silent, but the auxiliary verb that has moved to C is not:

(14) **Overt AUX moved to C shows no DFC effect**
Where [$_{C^o}$ will C] Mary — put the book –– ?

Similarly, when a main verb moves to C in a verb-second clause in a language such as Dutch or German, it is not silenced even though the complementizer is.

This observation suggests to us that the feature E that is obligatory in DFC environments is associated with the complementizer *qua* lexical item — that it is a property added to a complementizer when its features are being assembled in the *lexicon*, and is not the result of a later syntactic or phonological process. That is why the DFC effect is specific to the complementizer morpheme, and not to the contents of C as a whole.

This proposal suggests a new way of characterizing the DFC effect. The lexicon is also the place in the grammar where a head comes to bear those features that will dictate its properties as a probe for Agree and as a participant in Merge when that head is used in a syntactic derivation. A lexical item such as an interrogative complementizer that mandates the merger of a criterial *wh*-phrase as its specifier, for example, comes to bear this property when assembled in the lexicon, even though its effect will not be seen until it is employed in an actual derivation. For this reason, the DFC effect might not result from a silencing or deletion rule in the syntax (or syntax-sensitive phonology), as most proposals have assumed — but reflects a featural co-occurrence restriction on the building of complementizers in the lexicon. We thus propose (15) as the lexical rule that yields the DFC effect:

(15) **The DFC Rule: a featural co-occurrence restriction (final)**
A complementizer that bears a criterial merge feature F also bears E {obligatorily [English], optionally [Dutch]}. A complementizer that does not bear F does not bear E.

As we cautioned earlier, we will not go beyond this statement to offer an explanation for why this co-occurrence restriction exists in the first place. We leave that as a topic for future investigation.

We do leave open the possibility that other factors besides (15) may also yield a silent complementizer. Consider in this light the fact that the complementizer, but not the raised auxiliary verb, is silent in English matrix polarity questions that lack an overt *wh*-phrase in spec,CP, e.g. *will Mary put the book on the table?*. If the overtness of spec,CP is irrelevant to the DFC effect, as we have suggested above, then the proposal that matrix polarity questions involve a null *whether*, as proposed by Katz & Postal (1964, 95-97) and others, combined with Larson's (1985, 238) proposal that *whether* undergoes *wh*-movement to its surface position (cf. Wu 2022), does predict this result — and could be taken as a further argument that overtness is irrelevant to the DFC effect. There is some reason to think, however, that a rule distinct from (15) may mandate the silence of a complementizer in some constructions in some languages when head movement has taken place to C. In Dutch, for example, the complementizer is obligatorily silent whenever the finite verb has head-moved to C, even in dialects and registers that permit overt complementizers in DFC environments, tolerating examples like (11):

(16) **Complementizer obligatorily silent when finite V moves to C: Dutch (all dialects and registers)**
Wat heeft (*of/*dat) [$_{TP}$ mijn zuster ___ gelezen ___]]?
What has (*COMP) my sister read?
'What has my sister read.'

At the same time, some languages do permit or require the complementizer to remain overt in this environment as well.[7] We leave the distribution of these possibilities and the explanation for it for future work.

3 E-extension: from the DFC Rule to Sluicing

Consider now the phenomenon of sluicing in *wh*-questions, exemplified for English by (17), in which most of the clause except the *wh*-phrase is silenced. Sluicing is of course usually treated an example of a special phenomenon called *ellipsis*.

(17) **Sluicing**
a. Mary saw someone, but I don't know who.
b. "Max has invited someone." "Really, who?"

We begin by observing that sluicing obligatorily silences three things:

(i) the **complementizer**, and
(ii) any element **head-moved to C** (e.g. AUX in C), and
(iii) the **selected complement of C** (e.g. TP)

Properties (i) and (ii) taken together comprise Merchant's *Sluicing-COMP generalization*:

(18) **Sluicing-COMP generalization (adapted from Merchant 2001, 62)**
In sluicing, no overt material may appear in C.

As Merchant discusses, property (i) holds even when the language otherwise permits a *wh*-phrase to co-occur with an overt complementizer, as (19) shows. Property (ii) is illustrated by the English and Slovenian examples in (20).

(19) **Absence of obligatory DFC effect does not permit complementizer to survive sluicing (Dutch)**

a. **Absence of the Doubly-filled COMP Filter effect**
Ik weet niet, **wie (of)** hij gezien heeft
I know not **who if** he seen has
'I don't know who he has seen.'

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7. In Dinka, for example, a verb-second language, the verb moved to C co-occurs with an overt "declarative particle". As noted by Van Urk (2015, 102ff.), this particle agrees in φ-features with the nominal that occupies first position in the clause, suggesting that it is the complementizer that attracted this nominal in the first place. Similarly, in certain dialects of Dolomitic Ladin (Romance), as discussed by Hack (2014), *wh*-questions involve fronting of the finite verb to the immediate left of an interrogative particle *pa* or *po* — which might represent another instance of head-movement to C co-occuring with an overt complementizer. The Slavic interrogative particle *li*, found with somewhat varying distributions in the various languages of this family, may also instantiate this possibility.

b. **Complementizer must disappear under sluicing**
Hij heeft iemand gezien, maar ik weet niet **wie (*of)**.
he has someone seen, but I know not **who (*if)**
'He saw someone, but I don't know who.' (adapted from Merchant 2001, 74-75, exx. (97a), (99a-b))

(20) **No element moved to C survives sluicing**
a. **English AUX-to-C: AUX obligatorily silenced in sluicing**
A: Max has invited someone.
B: Really? Who (*has)? (i.e. *Who has ~~he invited?~~*)
b. **Slovenian clitic AUX-to-C: AUX obligatorily silenced in sluicing**
Špela je popravila nekako, a nisem vprašal, **kako (*je)**.
Špela AUX fixed something, but NEG.AUX.1SG asked **what AUX**
'Špela fixed something, but I didn't ask what.'
cf. *Špela je popravila nekako, a nisem vprašal, kako je popravila.*
what AUX *fixed*
'…*what she fixed*' (Merchant 2001, 66, ex. (84a))

If we think of the silencing characteristic of sluicing holistically, as a one-step ellipsis process, it looks like a rule specifically targeting C′. This looks odd to the syntactician, since we do not find single-bar projections specified as targets for movement or other well-established syntactic proceses (Landau 2020c, 378). Movement, for example, applies to heads and to maximal projections, not to intermediate projections. One might therefore try to imagine that the essential property of sluicing is the silencing of TP, licensed in some fashion by C or its specifier – with some other factor explaining the Sluicing-COMP generalization.[8] We wish to advance an alternative approach, which does silence a single-bar projection of C, like approaches that specifically mention C′, but roots this property in independent factors.[9]

8. See Landau (2020c, 376-9) for discussion of some of the alternatives that have been suggested, and the problems that they raise.

9. Landau (2020c, 375-378) offers an excellent summary of considerations that have been invoked in debating whether sluicing involves the silencing of C′ or just TP. Though he acknowledges that the C′-ellipsis view immediately accounts for the Sluicing-Comp generalization, while the TP-ellipsis view must seek other mechanisms, he nonetheless offers a scope argument based on properties of English T-to-C movement in favor of TP as the elided domain. The assumption that some additional factor can explain the Sluicing-Comp generalization is crucial to the argument — in particular, the ban on a raised auxiliary verb as a remnant in C when TP is elided. With this promissory note as background, Landau's argument rests on two claims about the scopal properties of modals when moved or not moved to C. First, he claims that a modal moved to C has its scope fixed in that position, and may not reconstruct to a lower position, e.g. below a clause-internal adverb. Second, he claims that the low scope that is disallowed for a modal moved to C in an unelided clause becomes possible if that clause that has undergone sluicing. He takes this as an argument that the modal never moved to C in the first place in constructions where sluicing has applied, a tenet that must be true if sluicing is TP-ellipsis.

We are not confident, however, that the baseline claim is correct — that a raised modal is fixed in scope and may not reconstruct below an adverb. Landau (2020c, 384, ex. (20)) offers examples like the following yes/no question as an initial illustration of this claim, with the judgments given:

(i) a. He can always be wrong.
always $\gg \Diamond$, $\Diamond \gg$ *always*
b. Can he always be wrong?
only $\Diamond \gg$ *always* (Landau's judgment)

We are not confident that reconstructed scope (always $\gg \Diamond$) is truly missing for the modal in (b). The following examples provide some context that we believe brings out the allegedly absent reconstructed reading:

(ii) a. Can I always sign up for health insurance (or do I need to wait for an enrollment period)?
(i.e. *Is it the case at all times that I can sign up…*)
b. Let me make sure I have correctly understood the nuances of church doctrine concerning papal infallibility. Can the Pope always be wrong except when he speaks *ex cathedra* — or are there other circumstances under which he is infallible?
(i.e. *Is it always the case that it is possible for the Pope to be wrong, except when…*)

Turning to *wh*-questions and interactions with sluicing, we continue to find reconstructed readings available for moved modals, with or without Sluicing. The parenthesized continuation helps to bring this reading out:

Sluicing affects a CP whose specifier is the final landing site for wh-movement, i.e. a “criterial position”. A CP does not permit sluicing if its specifier is merely an intermediate landing site for successive cyclic movement:

(21) **Intermediate landing sites of successive-cyclic movement do not license sluicing**

What was Mary worried [— that we might find out —]?

*And what was John worried [-- <*silenced material*>] ?

This is of course exactly the environment of the DFC effect. This observation suggests to us that the DFC Rule (15) is the actual motivating force behind sluicing — and that we should not think of sluicing holistically, but rather as arising in two steps. First, the complementizer of a clause comes to bear E in the lexicon as a consequence of the DFC rule. Optionally, as a second step that takes place in the syntax, a rule of **E-extension** may apply, which extends E from the complementizer just far enough to include the complementizer’s complement. This yields the silencing pattern that we have come to call sluicing.[10]

(22) **E-extension**

Optional: If H^o bears E, copy E to the smallest phrasal node containing H and its selected complement.

Because the story of sluicing begins with a DFC effect, property (i), the silencing of the complementizer, follows trivially. When E on a complementizer undergoes E-extension, the node to which E has been copied dominates not only the complement of C, but also any material that has head-moved to C, as well as the complement of C. This explains property (ii), and thus the Sluicing-COMP generalization in (18) — as well as the signature property of the

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(iii) *Speaker A*: You can always study most of the Romance languages at Harvard.
Speaker B:
a. Great! Which ones? (And which are only offered in certain semesters?)
b. Great! Which ones can I always study (and which are only offered in certain semesters)?

Landau’s discussion ranges over many kinds of modals and adverbs, and contains more detailed discussion than we can address here, so we acknowledge the need for further careful discussion. Note, however, that if one admits in the context of Landau’s proposals the possibility that silencing TP-internal material might eliminate the obligatoriness of T-to-C movement, one could in principle admit the same idea in the context of our own proposals presented below as well, providing a way of accomodating Landau’s factual claims without accepting his more general conclusions.

We are grateful to Idan Landau for discussing some of these issues with us.

10. An earlier proposal very similar to our rule of E-extension was advanced by Williams (1997, 621-627), who (like us) suggested that the silencing of a complement to a head is parasitic on the silencing of that head. Though Williams discussed several ellipsis constructions in this connection, he did not extend his proposal to Sluicing, and the overall context in which he made this proposal was different from ours. It is not clear that Sluicing behaves as predicted by the overall framework in which Williams’ rule played a role. Williams’ main concern was ”coordinate ellipsis” (i.e. gapping), which he analyzed as the obligatory extension of the“0” property (our “E”) of a verb in an initial conjunct to all elements of the verb’s complement that are not “disanaphoric” with their correspondents in the first conjunct. In the domain of coordinate ellipsis, this explains why *him* cannot corefer with *Bill* in a sentence like the one below under the analysis given — but can if *to him* is silenced as well.

(i) *John gave to Bill$_i$ a book today and 0_v to him$_i$ a record yesterday

Sluicing does not behave this way. The silencing of C does not entail that its complement must also be silenced unless disanaphoric. The full unsluiced version of the following example, though repetitive, is not disallowed:

(ii) John asked which book Mary$_i$ was reading, and Sue also asked which book (she$_i$ was reading).

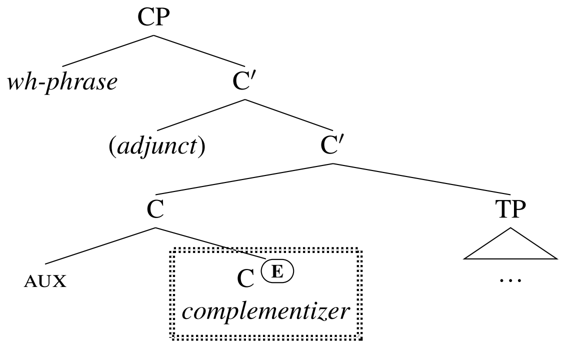
As is well-known, of course, a disanaphora effect does emerge in examples like (ii) if the process usually described as VP-ellipsis, rather than Sluicing, applies in the embedded clause – insofar as the string is ungrammatical unless the subject *she* receives the kind of prosodic prominence usually associated with focus and is interpreted as distinct in reference from the nominal in the antecedent clause with which it is parallel.

(iii) *John asked which book Mary$_i$ was reading and Sue also asked which book she$_i$ was.
[✓ if *she* bears a index j≠i and is stressed]

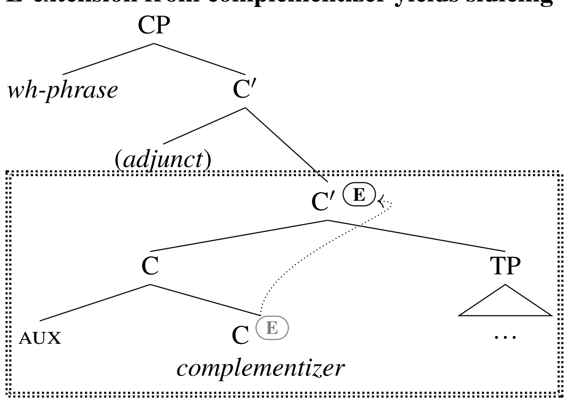
This effect is commonly attributed to a competition between Sluicing and VP Ellipsis such that the latter is blocked when the former is possible — the “MAXELIDE” constraint of Merchant (2008 [circulated 2001], 141). If MAXELIDE is what produces this effect, it supports the Uniformity of Silence thesis by treating Sluicing and verbal ellipsis phenomena as competitors. One might also ask whether Williams’ specific proposals might be unified with ours, suitably modified or extended, so that Williams’ proposals concerning disanaphora, rather than MAXELIDE might explain the facts. We leave this interesting possibility for future investigation. We are grateful to Edwin Williams (personal communication) for bringing his proposal to our attention.

construction, the silencing of the complementizer's selected complement. Because E-extension is limited to the minimal phrase containing the E-marked head and its complement, it will leave the specifier of CP unsilenced, yielding the characteristic look of sluicing constructions — but adjuncts in the C-system will also remain unsilenced, a property that we call **Adjunct Exclusion**. The effect of E-extension, including Adjunct Exclusion, is shown in (23) for constructions like those in (20) where an auxiliary verb has head-moved to C:

(23) a. **E assigned to complementizer by the DFC Rule**



b. **E-extension from complementizer yields sluicing**



The behavior of Slovenian discourse particles under sluicing, studied by Marušič et al. (2015) and Marušič et al. (2019), provides an example of Adjunct Exclusion, in this analysis of sluicing.[11] Slovenian interrogative discourse particles, which modify the discourse-level pragmatics of questions in various ways, follow the *wh*-phrase in a question but precede material known to occupy C — and survive sluicing. This is expected if these elements occupy a position within CP higher than the minimal C′ that contains just the contents of C and the complement of the complementizer, as shown in (23b).[12] The examples below show just two of a larger family of particles, with uniform syntactic behavior. The papers cited present the fuller set.

(24) **Slovenian discourse particles between Spec,CP and clitic AUX moved to C …**

a. Koga **pa** je Peter videl?
who.ACC *pa* AUX Peter.NOM saw
'Who did Peter see?'

b. Koga **že** je Peter videl?
who.ACC *že* AUX Peter.NOM saw
'Please remind me, who was it that Peter saw?'

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11. Marušič and colleagues presented examples like those as "a potential counter-example to Merchant's Sluicing-COMP generalization". It is therefore worth noting that our approach they actually follow from the same mechanism that otherwise derives this generalization.

12. Marušič et al. (2015, 55-58) provide several arguments that these particles are not internal to the *wh*-phrase.

(25) **…stranded by Sluicing**

a. Peter je videl nekoga. Koga **pa**?
Peter AUX saw someone. who.ACC *pa*
'Peter saw someone. Who <did he see>?'

b. Peter je videl nekoga. Koga **že**?
Peter AUX saw someone. who.ACC *že*
'Please remind me, who <did he see>?'

(adapted from Marušič et al. (2019, 196, ex. (5a-b)) and Marušič et al. (2015, 47-48, exx. (2a-b))

Our proposal predicts correctly that only when the complementizer bears E for other reasons will a sluicing configuration be possible (though, as discussed below, this is a necessary but not sufficient condition). If there are no independent processes that can suppress the pronunciation of sub-CP constituents that include the subject as sluicing does, the proposal also makes correct predictions about what will *not* be excluded from silencing in a sluicing construction. Sluicing-style silencing always affects the entire complement of a complementizer, and (ignoring for now the possibility that head-movement also yields a silenced complementizer) no part of that complement will fail to be silenced if any part of it has been.

Consider again in this light the Dutch dialects that permit the co-occurrence of overt complementizers with *wh*-phrases in spec,CP. The facts are somewhat more complex than those discussed above in connection with (11). As Merchant notes, many Dutch dialects and registers permit not just one but two overt complementizers following an overt wh-phrase (*of* 'if' followed by *dat* 'that'). Either or both of these complementizers may be missing in a question without sluicing, as shown in (26a). Under sluicing, however, both must be missing, as seen in (26b):

(26) a. **Two complementizers in Dutch…**
Ik weet niet, wie **(of)** **(dat)** hij gezien heeft.
I know not who if that he seen has
'I don't know who he has seen'

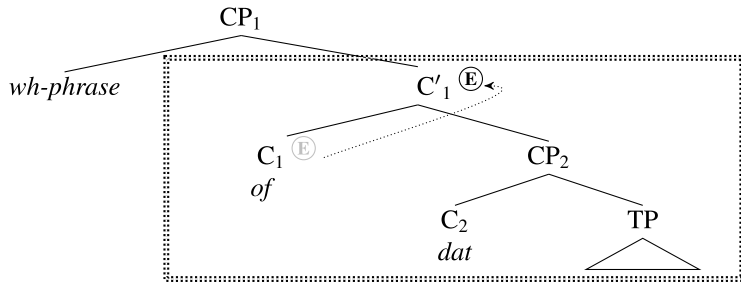
b. **…both obligatorily silent in sluicing**
Hij heeft iemand gezien, maar ik weet niet, wie (***of**) (***dat**).
He has someone seen but I know not who if that
'He has someone seen, but I don't know who.' (Merchant 2001, 74-75, exx. (97a), (99))

As the DFC Rule (15) states, the E-feature on a complementizer that takes a criterial specifier is optional in the relevant Dutch dialects. But only if the E-feature option is chosen will E-extension be able to apply, producing a sluicing configuration. We propose that the double-complementizer option in relevant Dutch dialects arises from an optionality in the subcategorization properties of the interrogative complementizer *of*, permitting it to optionally select a second CP headed by *dat* rather than TP (while *dat* itself selects only TP).[13] Both *Dat* and *of* may head an interrogative clause, which it does if and only if it is the highest complementizer of that clause and has been built in the lexicon with a criterial feature that attracts a *wh*-phrase.

We have three cases to consider. First, in a clause containing both *of* and *dat*, *of* must be the higher of the two (there are *of dat* clauses but no *dat of* clauses). Consequently, only *of* may receive an E-feature by the DFC Rule. If that E-feature is extended to include the *dat*-headed selected complement of *of*, *dat* will also be silenced, yielding a sluicing construction in which both complementizers are silenced (in addition to TP), as seen in (27).

13. Hoekstra & Zwart (1994) present several arguments against the alternative hypothesis that *of dat* is a single lexical item, rather than two separate heads. For example, the sequence *dat*+TP can be coordinated with another such sequence, with a shared *of*, in an embedded yes/no question.

(27) **E-extension in a two-complementizer interrogative clause silences both complementizers (Dutch)**



Second, in an interrogative clause that is headed by an instance of *of* that directly selects TP complement so that the clause entirely lacks *dat*, if *of* bears E and E-extension takes place, once again the result will contain no overt complementizer, since *of* is silenced and no *dat* was merged in the first place. Finally, if an interrogative clause is headed by *dat* and *of* has not been merged in the first place, once again E-extension will yield an outcome in which there is no overt complementizer. If no independent processes exist that can add E to various other portions of the structure (as we assume), the DFC Rule and E-extension together correctly predict Merchant's observation that the Sluicing-Comp Generalization extends to both complementizers in two-complementizer clauses.

4 Universals and limitations

Not every puzzling property of sluicing is predicted or explained by our proposal, unfortunately. For example, a strong and simple version of our proposal might predict that whenever C bears E, E-extension is possible, yielding sluicing. But despite the fact that languages like English do show DFC effects in relative clauses (as we saw in (4)), sluicing is not possible in a relative clause, as observed by Lobeck (1995, 55) and Merchant (2001, 58ff.):

(28) **No sluicing in a relative clause**

a. *I know a man whose daughters are linguists, and a woman whose sons. (i.e. whose sons are linguists)

b. *Show me the vowels over which you put diacritics in that language and I'll show you the ones under which. (i.e. under which you put diacritics in that language)

It thus must be possible to restrict E-extension so that it is prevented from applying to certain specific heads (such as the C that hosts relative clauses), and we will assume that heads can be marked with some feature that blocks E-extension in such cases. (We will need a very similar assumption for verbal ellipsis constructions in section 8 below.) Interestingly, however, the ban on sluicing in relative clauses appears to be extremely common cross-linguistically—perhaps universal (with apparent exceptions described as "rare" by Lipták & Aboh 2013, 105) — so one should obviously seek a deeper explanation for it than pure stipulation, which we cannot offer here.

On the other hand, the converse prediction might stand a chance of being true: that if a language permits a clausal ellipsis construction with the properties of sluicing discussed above (in particular, the limitation to criterial specifiers of CP and obedience to the Sluicing-comp generalization), the language permits or requires a silent complementizer independent of ellipsis. A potential counterexample would be a language with sluicing in which the complementizer that hosts a criterial *wh*-phrase is obligatorily overt when no ellipsis takes place, but the Sluicing-comp generalization is nonetheless obeyed, so that the complementizer is unpronounced only under sluicing. If a language with these properties is encountered, one might conclude that it poses a problem for our overall approach, but there are other possibilities as well. For example, one might propose that the language does make use of the DFC rule, and is fundamentally like colloquial French and Dutch in applying the DFC Rule only optionally — but with the additional property that once an E-feature is added to the complementizer, E-extension is obligatory. On this view, just as it is possible for a specific head to stipulate that E-extension is forbidden, it would be possible to stipulate that it is required.

Alternatively (and more interestingly), however, we might explore the possibility that what looks like an unsilenceable complementizer in non-ellipsis environments is not actually an instance of C after all — but rather a non-null distinct head (an affix) adjoined to C (perhaps thanks to movement from a lower position), with C itself E-marked and therefore null. If the E-feature on C is extended, it will silence the affix attached to it, just like the node labeled aux meets in (23b) and for the same reason.

We know of at least one language that presents exactly this puzzle, Tamazight Berber, which provides some evidence for the second approach just discussed. We are grateful to Hamid Ouali (p.c.) for the following observations. In this language, moved interrogative *wh*-phrases undergo *wh*-movement to a position in which they are obligatorily followed by a particle *ay* (with an allomorph *ag*).

(29) **Obligatory *ay* following moved interrogative *wh*-phrase [Tamazight Berber]**

a. mani-wa ***((a)g)** dda:n?
which-this AY left.PRT
'Which one left?'

b. mani-wa ***((a)g)** θ- li-θ
which-this AY 2-saw.SG
'Which one did he see?' (Hamid Ouali, p.c.)

Sluicing is possible in examples like these, but *ay* must be silent along with the rest of the clause under this circumstance:[14]

(30) **Overt *ay* impossible following moved interrogative *wh*-phrase in sluicing [Tamazight Berber]**

ʃa ħd i-dda walajnni ʃa-ur ssin-x mani-wa **(*ag)**
some one 3SG.M-left but NEG-NEG knew-1SG.M which-one **(*AY)**
'Someone left but I don't know which one.' (Hamid Ouali, p.c.)

If *ay* is an instance of C and heads its clause, the contrast between (29) and (30) looks like an instance of the Sluicing-COMP generalization in a language where C cannot otherwise be silent — a puzzle for our proposal of exactly the sort under discussion. It seems possible, however, that *ay* is actually a focus morpheme rather than a complementizer (perhaps moved from a lower focus head) — the second possibility considered above. As discussed by Ouali (2011, 71-72), *ay* is also used in what he calls a cleft construction, where it obligatorily precedes the focus element, and may introduce a relative clause (optionally, when a subject or object is relativized).

(31) ***Ay* in cleft and relative clause [Tamazight Berber]**

a. *cleft*
θamәttut-a **ay** θssәn Fatima
woman-this AY 2SG.know.PER Fatima
'It's this woman that Fatima knows.' (Ouali 2011, 71 ex. (153))

b. *relative clause*
θdda θmәttut **(ag)** snәn Fatima
3SG.F.leave.PER woman AY know.PER.PART Fatima
'The woman who knows Fatima has left.' (Ouali 2011, 69 ex. (145), corrected)

When a dative argument is fronted in any of these constructions, however *ay* (now obligatory in relative clauses) must co-occur prefixed to another element *mi*, as seen in (32) whose nature is a matter of disagreement in the literature, as discussed by Ouali (2011, 70):

(32) ***Ay+mi* with Ā-moved dative**

a. *wh-question*
m-**a(y)=mi** i-uʃa lktab?
WH-**AY-MI** 3SG-gave book
'To whom did he give the book?' (Ouali 2006, 123, *apud* Ouhalla & El Hankari 2015, ex (1))

14. As Hamid Ouali (p.c.) has pointed out to us, if the bare *wh*-element *ma* 'who' is substituted for the more clearly phrasal *wh*-phrase *mani-wa* 'which one', in (30), sluicing becomes impossible, with or without AY. If (30) had turned out to be similarly impossible, we might have been able to present Tamazight as a straightforward confirmation of the predictions of our overall hypothesis: a language with an obligatorily overt complementizer that lacks sluicing entirely. The reason an interrogative with *ma* actually does bar sluicing remains unknown. One might speculate some connection with D-linking (as suggested to us by Hamid Ouali), or alternatively, perhaps *ma* but not *mani-wa* behaves as a proclitic (and is thus blocked if E-extension silences the material that would otherwise follow it). We leave this question for future work.

b. *cleft*

θaməttut	**a(y)-mi**	θwʃa	Fatima	ssarut
woman	AY-MI	3SG.F.give.PER	Fatima	the.key

‘It’s this woman that Fatima gave the key to’

c. *relative clause*

θaməttut	**a(y)-mi**	θwʃa	Fatima	ləflus	θdda
woman	AY-MI	3SG.F.give.PER	Fatima	money	3SG.F.leave.PER

‘The woman that Fatima gave money to left.’ (Ouali 2011, 70 ex. (148), corrected)

One possibility considered reasonable by Ouali identifies *mi* and *ay* as distinct heads in the left periphery, forming a “split C structure” reminiscent of the Dutch constructions discussed above. Though the possibility Ouali entertains identifies *mi* as a lower head raised to *ay*, one might entertain the opposite possibility: that *ay* is the lower head and that *mi* is the higher head it raises to. On this view, *mi* would be the overt exponent of a normally silent complementizer that is pronounced as *mi* when its Ā-probe enters an Agree relation with a PP or oblique nominal. When the questioned, clefted, or relativized element is a PP, a prepositional morpheme intervenes between *ay* and *mi* — and indeed the dative examples above might actually contain an instance of dative *i* in that position as well (rendered obscure by the phonology of *ay*). One might view these prepositional morphemes as exponents of agreement. If so, Tamazight sluicing might conform to the predictions of our proposal after all, without the need to stipulate obligatory E-extension from a null complementizer.[15]

Our proposals do entail an unusual commitment, however. Phenomena analyzable as sluicing appear to be extraordinarily common in the languages of the world. Though we have not conducted a systematic investigation, it is our impression (supported by knowledgeable colleagues who we have consulted) that the following might be an absolute universal:[16]

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15. In the next section, we will argue at length that E-extension from the trace of a moved head is also possible (given the Uniformity of Silence Conjecture), and is the source of all phenomena traditionally described as VP-ellipsis. This proposal might offer an explanation for the behavior of constructions that look (at first) like sluicing in Gungbe, first discussed by Baltin (2010), building on Aboh (2004) — and superficially seem like counterexamples to the Sluicing-COMP generalization. In a Gungbe question, the fronted *wh*-phrase is followed by the particle *wὲ* (thus inviting an analysis as a complementizer), which is retained in what looks like sluicing:

(i)

Kòfí	ná	yrɔ́	mὲ	é	bɔ́	ùn	kànbíɔ́	ɔ́	mɛ́nù	**wὲ**
Kòfí	FUT	call	person	IND	but/and	I	ask	that	person.Q	FOC

‘Kofi will call someone and I wonder who.’ (Lipták & Aboh (2013, 103 ex.(4)))

There are good reasons, however, anticipated in the gloss above and discussed at length by Aboh (2004), for identifying *wὲ* as a focus head lower than C, rather than C itself — in which case it is no longer a counterexample to the Sluicing-COMP generalization on any analysis, as Baltin pointed out. Additionally, as discussed by Lipták & Aboh (2013), comparable ellipsis constructions are found in relative clauses, which is unexpected in a true sluicing construction. (They also argue that some examples that look like sluiced interrogatives are actually relative clauses with nominal heads, with ellipsis internal to the relative clause.) If *wὲ* raises to a higher head such as C, with its trace undergoing E-extension, these constructions fall together with those discussed in the following sections in which it is E-extension from a trace that produces the effect of ellipsis, rather than a complementizer that owes its E-feature to the DFC effect.

16. Overbold though we might be to propose (33) in the first place, we are also cautious in our formulation, since we have not investigated the sluicing-like phenomena known to exist in *wh*-in-situ languages such as Japanese (Takahashi 1994) or Turkish (Şener 2012; Ince 2012) from the standpoint of our proposal — nor languages like Hungarian in which which interrogative *wh*-movement targets positions lower than the specifier of CP. We leave these topics for future work, but do not intend our cautious formulation of the possible universal to suggest that it might not be extendable to wh-in-situ languages as well.

For example, perhaps a language without overt wh-movement but with sluicing does permit overt wh-movement after all — and shows some version of DFC effects — but with a stipulation that E-extension is *obligatory* from null interrogative C. This would entail that overt movement must always be accompanied by sluicing. In a Japanese setting, such a proposal would require that the interrogative marker *ka* be understood as a Q-morpheme moved to a position within CP outside the minimal C′ (Watanabe 1992; Hagstrom 1998, 40ff.; Cable 2010, 84ff.) or else as a higher head of some sort — and not as C itself, given its obligatory overt appearance in sluicing contexts:

(33) **Conjecture: the universality of sluicing**
If a language has overt *wh*-movement to a clausal left-peripheral position, it permits sluicing.

The universality or near-universality of sluicing entails a similar ubiquity for the DFC Rule (in its optional or obligatory variant), at least among languages with overt *wh*-movement. It remains to be seen whether this proposal is sustainable.[17]

Part II: Verb-headed constituent ellipsis

5 Overview: silenced traces and E-extension

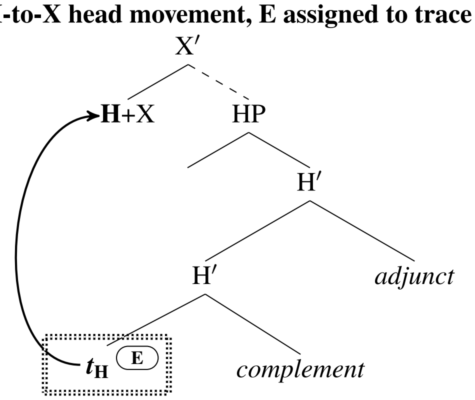
In the first part of this paper, we put forward the Uniformity of Silencing conjecture, repeated below, according to which all instances of silencing have the same source: a feature that we call E. We then posited an optional rule of E-extension to derive sluicing from the silencing of complementizers by the DFC rule.

(2) **Uniformity of Silencing Conjecture (repeated)**
A phrase or lexical item is silent if and only if it bears the feature E.

Uniformity of silence has an implication for movement, because if silencing is always the result of E-assignment, then unpronounced copies in movement chains must also be assigned E, in accordance with our conjecture. This, in principle, should be true of head movement as well: When a head H moves, its lower copy is silenced in virtue of it being assigned E, as in (34).

This raises an immediate question: can the optional rule of **E-extension** apply to the E-feature assigned to a trace of head movement, as we argued is the case for the E-feature on complementizers? If the E-feature on a head H can extend, the result should constitute ellipsis of the minimal H-bar projection containing both the trace of H and its complement, as in (35).

(34) **H-to-X head movement, E assigned to trace**



(i) Mary-ga nanika-o katta rasii ga, boku-wa [nani-o ka] wakaranai.
Mary-NOM something-ACC bought likely but I-TOP [what-ACC KA] not.know
'It is likely Mary bought something, but I don't know what.' (Takahashi 1994, 266, ex.(3))

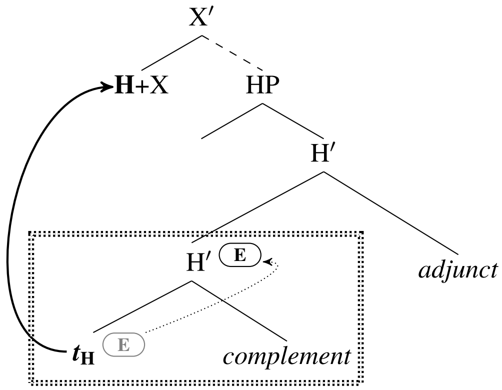
A similar entailment between properties of C and obligatory sluicing might be at stake in the obligatory link between so-called "Swiping" (optional inversion of a preposition with a monosyllabic *wh*-phrase; Ross 1969b, 265, Merchant 2001, 64) and Sluicing in languages like English:

(ii) Mary is collaborating with someone on this, but I don't actually know who with (*she is collaborating).

Perhaps Swiping is triggered by a flavor of null interrogative C capable of triggering subextraction from its own specifier (Richards 2004), but mandating E-extension.

17. An even more ambitious version of this commitment is Koopman's (2000, 323 ff.) conjecture that phenomena analogous to the DFC effect can be found universally in every syntactic projection — which raises questions in the present context concerning whether ellipsis phenomena comparable to sluicing can be similarly motivated in other projections.

(35) **H-to-X head movement, E assigned to trace, E extended to H′**



In this section we propose that **head-stranding minimal head-bar ellipsis** is indeed the correct description of what takes place in constructions traditionally analyzed as involving verb-stranding VP-ellipsis (VVPE). Just as with sluicing, we predict that the size of the elided constituent is restricted by the identity of the E-bearing head, whose E-feature extends to yield further silencing. Crucially, it is only the minimal bar-projection of that head that can be elided, resulting in an ellipsis site that includes the head's complement — but excludes both adjuncts to H and all HP-external material.

We will argue that this phenomenon is exactly what Landau has discussed under the rubric of **Adjunct Exclusion** in configurations previously analyzed as VVPE (Landau 2018, 2020a, 2022; and work on which he builds). While Landau took these observations to constitute an argument against VVPE, and in favor of *argument ellipsis* (henceforth, AE) as the analysis of object gaps in these constructions, we will argue that they actually favor a verb-stranding ellipsis analysis — unifying the behavior of the trace of V-movement with the behavior of silenced complementizers in sluicing configurations. Ultimately, in the final sections of this paper, we will advance the strongest version of our proposal and argue that *all* so-called VP-ellipsis (including ellipsis of phrases headed by an auxiliary verb) is actually Verb-stranding Minimal V′ Ellipsis (**VMV′E**) — always fed by movement of the rightmost verb that survives ellipsis and the concomitant obligatory silencing of its trace. This proposal dovetails with proposals by Thoms (2010) and Harwood (2014), among others, which also posit verb movement as a necessary precursor to "VP ellipsis" (in somewhat different ways). Some of their evidence will be adduced to support our proposal as well.

Before we continue, however, it is important to highlight what our argument is *not*. First, we assume that adjunct exclusion is a real phenomenon in the ellipsis constructions we discuss here, and will not occupy ourselves with a defense of its existence. That empirical claim has been called into question in a recent exchange between Landau and Simpson (Simpson 2023; Landau 2023; cf. Merchant 2018, 262-263, whose position is similar to Simpson's, with evidence from Greek) — and we regard Landau's arguments in favor of the claim as successful. In agreement with Landau, we will thus take it for granted that adjunct exclusion does exist and focus our attention on the explanation for the phenomenon, an area in which we will disagree with Landau's conclusions. At the same time, however, we are not logically compelled to take issue with Landau's claim that AE is a real phenomenon that offers a viable analysis for certain object gaps, particularly in the Hebrew VP domain. AE may exist as a possibility alongside verb-stranding verbal projection ellipsis, though we are unsure whether this possibility makes testable predictions distinct from those made if AE does not exist alongside our proposals.[18] What is important to us here is the fact that even if AE does turn out to be possible in languages like Hebrew, head-stranding minimal head-bar ellipsis also exists, and must be posited to explain phenomena that AE alone cannot.[19] At the same time, in the domain of verbal ellipsis phenomena, we will disagree not only with Landau's proposal that fails to posit verb-stranding verb-projection ellipsis in the first place, but also with earlier standard proposals that countenance verb-stranding ellipsis but posit a full VP as the elided constituent.

18. If AE does exist, it would be interesting, of course, to investigate whether it also arises from E-extension due to head-movement of some element within DP.

19. Of course, if AE is real, and involves a silent phrase such as a D with an NP complement, and not just a single E-marked lexical item of category D, we will need to ask whether the silencing of the NP portion of the phrase falls together with other phenomena viewed as instances of "ellipsis" (Elbourne 2001, 2005) and therefore should be attributed to E-extension.

In the sections that follow, focusing on Landau's examples from Hebrew, we first show how adjunct exclusion is accounted for by a theory of ellipsis as E-extension. We then present previously unnoticed observations from both Hebrew and English to illustrate that adjunct exclusion is just one instantiation of a broader phenomenon, *generalized higher-element exclusion* – as predicted by our theory of ellipsis. Finally, note that we engage with a series of arguments posited by Landau *against* the existence of verb-stranding verbal-projection ellipsis in an appendix — addressing some of his data, while acknowledging some data that our proposal does not currently accommodate.

6 Adjunct exclusion phenomena in verb-headed constituent ellipsis

The sentence in (36) contains an example of an object gap in Hebrew. Its second conjunct conveys the information that the speaker thinks Yossi knows Dina, despite the fact that the object of the verb *know* in that conjunct is null. A commonly proposed analysis of such examples (and of Hebrew object gaps more generally) since Doron (1999) and Goldberg (2005) posits extraction of the verb from VP followed by ellipsis of the remnant VP, which silences the direct object and any other elements remaining within VP at that point — while the verb itself remains overt. This is the derivation referred to as VVPE, whose existence is challenged by Landau (2018), schematized in (37).

(36) **Hebrew object-gap construction (standard analysis: VVPE)**

ani	makir	et	Dina	ve-	ani	xošev	še-	yossi	**gam**	**makir**
I	know.PRS.1MSG	ACC	Dina	and	I	think.PRS.1MSG	that-	Yossi	also	know.PRS.3MSG

'I know Dina, and I think that Yossi also knows her.'

(37) **VVPE analysis of (36)**

***v*-to-T movement**

... *Yosi also* ***knows*** +T $\underbrace{[_{vP}\ \_\_\_\ \textit{her}]}_{\textbf{elided}}$

Landau's argument against this approach is simple (adapting an earlier argument for Japanese by Oku 1998, 304-305): If it is indeed VVPE that generates Hebrew object gaps, and the verb itself is overt in virtue of it having moved out of the VP prior to ellipsis, then we should expect other VP-internal material that has not moved to elide together with the verbal object. Strikingly, this prediction is not borne out. If the antecedent VP contains an adjunct, that adjunct may *not* form part of the ellipsis site. The examples in (38) illustrate this observation:

(38) **Adjunct exclusion under VP ellipsis (Hebrew)**

Q:

pagašta	et	John	be-Pariz?
met.2SG.M	ACC	John	in-Paris

'Did you meet John in Paris?'

A: #

lo	pagašti.	ze	haya	be-London
NEG	met.1SG.	it	was	in-London

'I didn't meet him. It was in London.'

A′: ✓

lo.	ze	haya	be-London
no.	it	was	in-London

'No. It was in London.' (Landau 2023, ex. 12)

In Hebrew, a question like *did you meet John in Paris?* can be answered negatively either by using a sentence with an object gap, like *lo pagašti* 'didn't meet' in (38-A) — or with a simple 'no' answer as in (38-A′). The two possibilities are not equivalent, however. Negation in the simple *no*-answer as in (38-A′) can be understood as targeting either the entire VP containing the adjunct 'in Paris', or a sub-constituent containing just the verb and its object, i.e., 'meet John'. Because of this ambiguity, a simple 'no' answer can be understood in two ways: (i) as denying that the speaker met John at all; or (ii) as merely denying that a meeting took place in Paris. Possibility (ii) permits the continuation 'it was in London'. The answer *lo pagašti* 'didn't meet', on the other hand, can only be understood as an instance of (i), denying that the speaker met John at all — and not as an instance of (ii), as illustrated by the fact that the speaker cannot continue this answer by saying that she did meet John, but in London. Additional examples making the same point can be seen in (39).

(39) **Adjunct exclusion under VP ellipsis: additional examples (Hebrew)**

a. **Q:** ata makir ota me-ha-tixon?
you know.PRS.2MSG her from-the-high.school
'Do you know her from high school?'

A: # lo makir. ani makir ota me-ha-universita
NEG know.PRS2MSG. I know her from-the-university
'I don't know her. I know her from the university.'

A′: ✓ lo. ani makir ota me-ha-universita
no. I know her from-the-university
'No. I know her from the university.'

(adapted from Landau 2018, following the exchange in Simpson 2023 and Landau 2023)

b. **Q:** Yosi afa et ha-uga lefi ha-matkon. hi hayta me'ula.
Yosi baked ACC the-cake by the-recipe. it was fabulous.
'Yosi baked the cake according to the recipe. It was fabulous.'

A: Gil lo afa. #hi hayta mag'ila
Gill not baked. It was gross.
'Gil didn't bake. #It was gross'

A′: Gil, lo. hi hayta mag'ila.
Gil, not. it was gross
'Gil didn't. It was gross.' (Landau 2018, 21-22, exx. (38)-(39))

The general argument against VVPE advanced by Landau that is supported by examples like (38) and (39a-b) is the following. We can diagnose the presence of an adjunct in a VP-ellipsis site by examining whether a negative particle external to the ellipsis site can be used to deny only the truth-conditional contribution of the adjunct. If Hebrew clauses with object gaps are indeed derived by VP-ellipsis, and if negation outside a VP-ellipsis site may (at least optionally) be used to deny only the truth-conditional contribution of an adjunct, then if negation outscoping an object gap is unable to do so, these object gaps cannot be the result of VP-ellipsis. In English constructions traditionally analyzed as pure VP-ellipsis, by contrast, negation outside a VP-ellipsis site may target a VP-internal adjunct alone:

(40) **Negation outside VP-ellipsis targets adjunct**

Q: Did you meet John in Paris?

A: I didn't. I met him in London.

This argument goes through, however, only if we assume that what is elided after a verbal head raises in Hebrew object gaps is the entire maximal projection of the raised head. In our discussion of sluicing, we proposed a view of ellipsis that does not involve the silencing of entire maximal projections. Instead, we proposed that ellipsis arises when the silencing of a head is extended by the optional rule of E-extension stated in (22), repeated below, which silences the minimal projection containing the silenced head and its complement.

(41) **E-extension**

Optional: If H° bears E, copy E to the smallest phrasal node containing H and its selected complement.

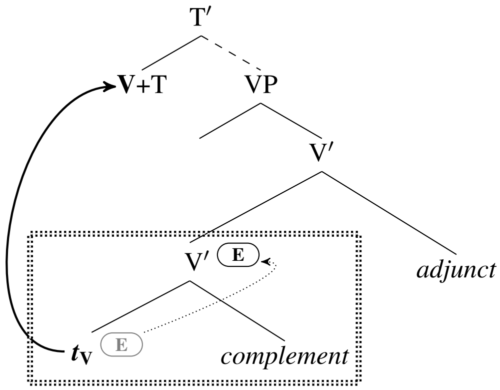
In that section, we analyzed sluicing as the consequence of E-extension from a head assigned E as an instance of the DFC effect. We now propose that lower copies of heads that have undergone movement are also unpronounced as a result of the assignment of E to the position that they occupy — and that E-extension may apply there as well.

Suppose now that Hebrew object gaps in examples like (38) and (39a-b) are derived by verb movement followed by E-extension from the raised verb's lower copy. Holding constant the statement of E-extension motivated by the facts of sluicing, E-extension will automatically exclude any VP-internal adjuncts from the silenced material. Crucially, the resulting ellipsis site[20] will be exactly as big as the smallest constituent containing the trace of the raised

20. We will continue to use the term "ellipsis site" to describe phrases silenced by E-extension, because it is familiar — though we posit no notion of ellipsis independent from E-extension.

verb and its selected complements, as illustrated in (42). We will refer to such configurations as instances of *verb-stranding minimal V′ ellipsis* (**VMV′E**).

(42) **Verb-stranding minimal V′ ellipsis (VMV′E)**



The survival of overt VP-internal adjuncts in such constructions thus follows from the same property of E-extension that permitted the Slovenian discourse particles in (25) to remain pronounced in sluicing structures — and therefore provides an argument for the unified theory we are proposing. And indeed, when an excluded adjunct is pronounced in a negated verb-echo answer as in (43-A), the result is grammatical and felicitous, as now negation can target the truth-conditional contribution of the adjunct.

(43) **Inclusion of overt adjuncts above a silenced V′ (Hebrew)**

Q: pagašta et John be-Pariz?
met.2SG.M ACC John in-Paris
'Did you meet John in Paris?'

A: lo pagašti be-Pariz. ze haya be-London
NEG met.1SG in-Paris. it was in-London
'I didn't meet him in Paris. It was in London.'

Furthermore, we predict not only exclusion of adjuncts, but also exclusion of any element higher than the complement of the moved head's trace. When a head H raises, because E-extension silences only the minimal H′ containing H and its complements, nothing else on the path between the original position of H and its new position will be silenced. This includes specifiers and adjuncts of H as well as material entirely external to HP, the maximal projection of the moved head. We call this prediction *higher-element exclusion*, and show that it is borne out in the next section. We also show that another, related prediction holds; the phenomenon of adjunct exclusion is predicted to be fully general in the sense that it should not be restricted to VP-adjuncts in particular. Rather, silencing triggered by movement of any head H along the clausal spine should result in an ellipsis site that excludes any adjuncts of H.

This final prediction provides us with a segue to the final sections of this paper, in which we will argue that *all* VP-ellipsis phenomena are derived via E-extension of an independently assigned E-feature on a silenced head. This means that every construction traditionally analyzed as VP-ellipsis actually involves a VMV′E construction, derived by movement of the rightmost verb that survives the silencing of its smallest projection. Our most substantial argument in favor of this approach will arise as part of a broader discussion of AuxP ellipsis in the English auxiliary system, but the logic of both *higher-element exclusion* and *generalized adjunct exclusion* also requires this conclusion by requiring the non-existence of any kind of maximal-projection ellipsis in the verbal domain.

7 Higher-element exclusion

Sentence negation exclusion. The work of Emonds (1976, 213-216), Pollock (1989, 366-370), and others has shown that English auxiliary verbs (and their counterparts in many languages) are initially merged below sentence negation but may raise above it thanks to AUX-to-T movement that crosses negation. When this happens, the lower copy of head-movement, located below negation, will be assigned an E-feature. This E-feature, if extended, will create a configuration of the sort traditionally analyzed as ellipsis in the verbal domain. By hypothesis, such movement also takes place in the absence of overt elements like sentence negation intervening between the initial and final positions of the raised auxiliary verb, with E-extension from the auxiliary verb's trace yielding familiar examples like (44) — on our approach, just another instance of VMV′E.

(44) **Aux-to-T movement + E-extension → ellipsis construction**

aux-to-T movement

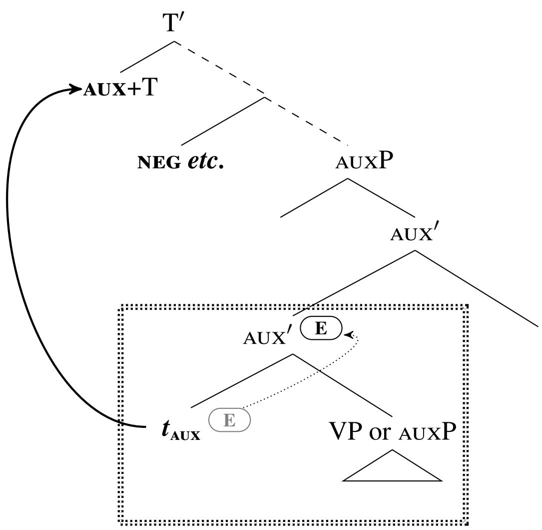
(Mary has left the room, and) *John* ***has*** +T [___ *left the room*] *too*

elided

In discussing such examples, it will be convenient to have a name for the element that most immediately precedes the silenced verbal phrase in such constructions. We will use the term *right-survivor* for this element. The auxiliary verb *has* in (44) is thus a right-survivor.

Our proposal now makes a prediction concerning situations in which an auxiliary verb moves across sentence negation. Any ellipsis configuration attributable to auxiliary-verb movement across negation followed by E-extension from the trace will silence the complement of the head in its original position — but nothing else. As a consequence, the negation that it crossed will not be silenced, and must be a right-survivor. The prediction is illustrated in (45).

(45) **E-extension from the trace of aux-to-T movement**



The prediction is borne out: sentence negation is not elideable in aux-stranding ellipsis, as illustrated in (46), but may survive as the remnant of ellipsis triggered by extension of an E-feature on the trace of aux-movement (47).

(46) **Sentence negation not elidable in aux-stranding ellipsis**

a. John has not left the room, #and Mary has too.
Intended meaning: *Mary has not left the room too.*

b. Sue is not doing the homework, #and Mary is either.
Intended meaning: *Mary is not doing the homework either.*

c. Our party will not win this election, #and their party also will.
Intended meaning: *Their party also will not win this election.*

(47) **Sentence negation may be a remnant**

a. John has not left the room, and Mary hasn't either/?too.

b. Sue is not doing the homework, and Mary is not either.

c. Our party will not win this election, and their party also won't.

This contrast cannot be attributed to a general ban on the silencing of negation or downward entailing operators. Constituent negation (which resides lower in the structure than sentential negation) is elideable, as are other negative elements located below the auxiliary verb that survives ellipsis – as illustrated in (48)-(49).

(48) **Lower (constituent) negation is elidable → no general ban on silencing negation**
a. Sue has been not doing the homework lately, and John has (been) too.
b. You could have simply not tackled that problem on the exam, and Bill could (have) too.

(49) **Negative elements other than sentence negation are elidable, unlike sentence negation**
a. Sue was unable to speak French, and Mary was too.
compare: Sue could not speak French, #and Mary could too/either.
b. Sue has finished none of the homework problems, and Mary has too.
compare: Sue has not finished any of the homework problems, #and Mary has too/either.

The observations in (46)-(47) are also not unique to elements functioning as auxiliary verbs, but extend to copular verb *be* (which raises to T just like its auxiliary-verb counterpart).[21]

(50) **Copular be behaves similarly**
You, Mary, are not a bad person. #And you, Tom, also are. (vs. ✓And you, Tom, also are not.)
(cf. *The same is true of you, Tom.*)

The fact that sentence negation may not be elided under a stranded verb that has raised to T follows from our proposals, which limit ellipsis to the minimal phrase that includes both the trace of the raised auxiliary and its complement. The unification of this phenomenon with Adjunct Exclusion made possible by our proposal provides an argument its favor.

Other instances of higher-element exclusion. The important property of sentence negation that leads to its exclusion from ellipsis in examples like those studied in the previous subsection is its location: above the initial position of the verb that raised to T, but below its final position. The theory thus predicts that *any* element located within this region will be excluded from ellipsis in the same manner. This prediction is confirmed by the behavior of high adverbial 'only' in Hebrew, whose interaction with ellipsis was first explored by Benbaji (2022), and which we will use here in further motivating our proposal.

Example (51-Q) contains an adverbial use of *rak* 'only' that associates with the lower focused nominal 'wine'. Movement of the main verb above *rak* accompanied by E-extension from its trace cannot silence *rak*, since the ellipsis site is limited to the verb's subcategorized complement of the verb, and *rak* is outside that domain. The bare-verb answer (51-A) is thus illegitimate as an affirmative response to the question 'Did Dina bring only WINE' — though it could be an affirmative response to the simpler question 'Did Dina bring wine?', omitting *only*.

(51) **No verb-echo answers to polar questions with adverbial *only***
Q. Dina rak hevia jain?
Dina only bring.PST.3FSG wine
'Did Dina bring only WINE?'
A. #hevia / ✓ ken
bring.PST.3FSG / yes
Intended: 'yes'

As Benbaji (2022) notes, if the silenced constituent below the verb *bring* in (51-A) must be parsed as lacking a silenced instance of *rak*, the infelicity of the bare-verb answer straightforwardly follows. That answer simply conveys that Dina brought wine. The question to which it is a response, however, already presupposed that Dina brought wine — since *only* introduces a presupposition that its prejacent is true (Horn, 1969, Karttunen & Peters, 1979). Thus, if the answer cannot be understood as containing 'only' as part of its ellipsis site, it is simply repeating what the question already presupposes (that Dina brought wine) and is thus correctly predicted to be infelicitous.

The constraint seen in (51) is not limited to affirmative answers, but is a general fact about ellipsis configurations made possible by verb raising and E-extension. The short verb raising already discussed and illustrated in examples like (36) shows the same pattern.

21. Non-perfect uses of *have* that optionally raise to T behave the same way, as predicted: *Sue had not the slightest intention of studying syntax — #and John had too/either.* In this case, however, the alternative that preserves *not* unsilenced is also unacceptable: *Sue had not the slightest intention of studying syntax — and John had not too/either*, for unknown reasons.

(52) a. Dina rak hevia jain ve- Ben gam hevi.
Dina only bring.PST.3FSG wine and- Ben too bring.PST.3MSG
'Dina only brought WINE and Ben brought wine too.'
(Felicitous if Ben brought both wine and beer!)

b. Dina rak hevia jain ve- Ben lo hevi.
Dina only bring.PST.3FSG wine and- Ben NEG bring.PST.3MSG
'Dina only brought WINE and Ben didn't bring wine.'
(Infelicitous if Ben brought both wine and beer!)

At this point, however, we might expect that the bare-verb answer in (51) could be rendered felicitous if *rak* remains unelided as a right-survivor. In that sense, Hebrew *rak* contrasts not only with English sentential negation, which cannot form part of the ellipsis site but could be a right survivor, but also with other Hebrew adjuncts that can optionally survive ellipsis (43).

(53) ***Rak* as remnant does not improve (51)**

Q. Dina rak hevia jain?
Dina only bring.PST.3FSG wine
'Did Dina bring only WINE?'

A. *hevia rak
bring.PST.3FSG only
Intended: 'yes'

As it happens, however, the impossibility of *rak* as a remnant may be attributed to an independently motivated constraint on association with focus across an ellipsis boundary, discovered by Beaver & Clark (2008):

(54) **Beaver-Clark constraint (Beaver & Clark 2002, 2003, 2008)**
Only outside a silenced domain may not associate with a focused element within that domain.[22]

The effect of the Beaver-Clark constraint can be seen in both VP-ellipsis constructions and sluicing. This supports the claim that the constraint is general and real, and not limited to the specific configuration under discussion in this section. Its effect can be seen in English examples like (55):

(55) **Examples of the Beaver-Clark constraint**

a. **Verbal ellipsis**
John only eats CHEESE. *Bill only does too. <elided: *eat cheese*>
(cf. John only eats CHEESE. Bill does too. <elided: *only eat cheese*>) (Stockwell 2020, 245)

b. **Sluicing**
Sue and Tom already know why everyone else left the concert early. Consequently ...
*Sue only wants to know why BILL left the concert early, and Tom only wants to know why too.[23]

Note that the availability of the alternate version of (55a) in which *only* forms part of the ellipsis site contrasts with the bare-verb answer in (51-A), which does not permit a comparable parse in which *rak* forms part of the ellipsis site. This indicates that Hebrew adverbial *rak* must occupy a position higher than the base position of the raised

22. For the purposes of the discussion in this section, we can treat this Beaver-Clark as a primitive constraint on focus association across an ellipsis boundary. However, our use of the constraint can be made compatible with its reduction to the interaction of competing requirements on pitch accent, suggested in Merchant 2018.

23. Crucially, a baseline example omitting 'only' is fully acceptable:

(i) Sue wants to know why BILL left the concert early, and Tom wants to know why too.

Comparable baseline examples with *wh*-phrases other than *why* are not acceptable, in a manner reminiscent of the effects of the MaxElide constraint proposed by Merchant (2008 [circulated 2001], 141), unless the *wh*-phrases differ and contrast.

(ii) a. *Sue wants to know which concerts BILL left early, and Tom wants to know which concerts too.
b. Sue wants to know which concerts BILL left early, and Tom wants to know which plays.

We do not know why *why*-questions are not subject to this obligatory contrast requirement. Though an additional Beaver-Clark effect is detectable if we add *only* to (ii-b), it is hard to ensure that this is not due to difficulties associating *only* with a focused but silenced occurrence of *Bill* rather than the unsilenced focused *wh*-phrase that intervenes.

main verb, while English *only* may occupy a position below the base position of an auxiliary verb.[24] We return to the analysis of English auxiliaries below, noting only for now that if our overall proposal is correct, the very fact that an ellipsis site may follow an auxiliary verb such as *do* entails that the auxiliary has raised to T from a lower position, an observation we shall explore in section 9.

Hebrew, like English, may also place *rak* adjacent to the focused element itself, so that in a sentence meaning 'Did Dina bring wine only to Dan?', for example, *rak* may be adnominal.[25] Surprisingly, perhaps, these constructions interact with ellipsis just like their counterparts in which *rak* appears in an adverbial position — despite the fact that the overt occurrence of *rak* in the question could presumably form part of an ellipsis site in the answer that also includes the focused nominal.

(56) **No verb-echo answers to polar questions with adnominal *only***

Q. Dina hevia jain rak le- Dan?
Dina bring.PST.3FSG wine only to- Dan
'Did Dina bring WINE only to Dan?'

A. #hevia / ✓ ken
bring.PST.3FSG / yes
Intended: 'yes'

In fact, however, the unacceptability of such examples can be explained in the same way we have explained its predecessors — as a higher-element exclusion effect interacting with the Beaver-Clark constraint — so long as we accept Hirsch's (2017, 234ff.) proposals concerning the syntax of the positional alternations of *only*. Hirsch argues that in an 'only' construction, no matter where we hear the word 'only', there is always an adverbial instance of *only* merged in an adjunct position above vP or a higher node.[26] The overt instance of *only* that appears adjacent to its focus associate (as in *Dina brought wine only to Dan*) is actually a realization of focus whose morphology reflects agreement with the silent adverbial 'only'. It is the silent adverbial operator that does the real work of supplying the meaning of *only* in such constructions. Thus, the sentence *Dina brought wine only to Dan*, consists of the material in (57), where $\varnothing_{\text{ONLY}}$ is the operator contributing *only*'s meaning, and the string *only* that surfaces in the position linearly adjacent to the focus associate is an agreement reflex, which we gloss as FOC.[27]

(57) ***Only* is always adverbial**

Dina $\varnothing_{\textbf{ONLY}}$ brought only only to Dan.
Dina **only** *brought wine* **FOC** *to Dan*

Above, we illustrated the Beaver-Clark constraint (55) with examples that included an overt adverbial *only* coupled with a focused associate whose focus property was realized only prosodically. All things being equal, however, the Beaver-Clark constraint should also hold when the silent version of adverbial *only* is employed and focus on its associate would have been realized as overt *only*, had E-extension (i.e. an ellipsis configuration) not silenced it. If the

24. The order **rak hevia* is also impossible, though this could be due not only to the Beaver-Clark constraint, but also to the absence of a position for *rak* above the landing site of the moved verb (and no way for it to have raised from a lower position along with the verb).

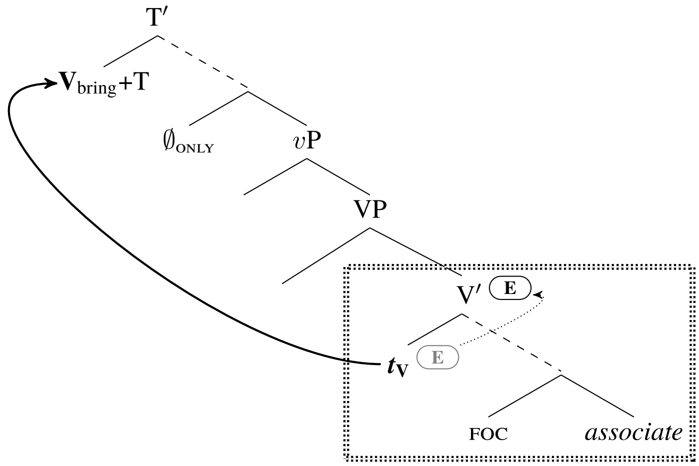
25. We use a ditransitive verb here to permit the placement of *rak* at some distance from the verb, to ensure that it is not adverbial, surfacing in a seemingly adnominal position due to verb movement over it.

26. vP is arguably the smallest node in the derivation whose semantic denotation is of a propositional type, and given that *only* is a function from propositions to propositions, Hirsch (2017) assumes that at the very least, vP has to be in the scope of adverbial *only*.

27. Only one instance of the form 'only' (either the adverb or the agreeing focus particle) may be phonologically realized at a time in English and Hebrew. See Hirsch (2017, 50) for a formal statement of the rules necessary to regulate this. As discussed by Yip (2024), however, some languages do not have this limitation, and permit the simultaneous expression of both the adverbial and the focus element.

analysis of an example like the bare-verb answer in (56-A) is (58) below, it is correctly ruled out by the Beaver-Clark constraint (54).[28]

(58) **VMV′E with a silent *only***



If our view of the phenomena discussed in this section is correct, it thus provides independent evidence for Hirsch's analysis of the alternations characteristic of *only*, as well as a wider sphere of action for the Beaver-Clark constraint than previously suspected.[29]

The strongly positional nature of our account of exclusion phenomena makes other testable predictions about Hebrew 'only', to which we now turn. As we have seen, whenever an element such as negation or Hebrew adverbial 'only' lies outside the minimal phrasal projection of the original position of a moved verb (main or auxiliary), E-extension from the trace of that verb will not silence it (potentially producing a violation of the Beaver-Clark constraint). If the base position of the moving verb is *higher* than negation or adverbial 'only', however, so that it is included in the verb's subcategorized complement, no problem should arise: no exclusion effect nor Beaver-Clark violation. Hebrew 'only' provides an example of this as well, when we examine its interaction with auxiliary verbs occupying high vs. low base positions.

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28. Benbaji (2022) used the examples in (51) and (56) as evidence against a VVPE account of Hebrew object gaps and in favor of an AE account, following Landau (2018, 2020a) — arguing that the absence of verb-echo answers to questions with *only* cannot be derived as a violation of Beaver and Clark's generalization if these answers are derived as VVPE in the traditional sense. In that case, nothing would prevent a derivation along the lines of (58) in which the ellipsis site is big enough to contain both covert *only* and its focus associate. However, Benbaji's arguments against VVPE do not extend to VMV′E for which we advocate here. On our proposal, the size of ellipsis is constrained by the original position of the stranded verb, and therefore prevents the ellipsis site from being large enough to contain covert *only*.

29. Of course, the Beaver-Clark constraint, which we have just stipulated as a fact, should ultimately be explained as a consequence of more general considerations. One promising approach is suggested by Merchant (2018). Building on earlier ideas by Tancredi (1992), Merchant suggests, in effect, that the Beaver-Clark constraint results from the interaction of two independent facts about pitch accent:

I. A focus sensitive operator like *only* demands that its associate be focused-marked, and a focus-marked constituent must show prominence by bearing pitch accent.

II. Ellipsis, viewed as a species of *de-accenting*, effectively removes the pitch accent of anything in the ellipsis site.

It follows from the claim in II that when the associate of *only* is embedded in an ellipsis site it cannot possibly bear pitch accent, which immediately conflicts with the requirement stated in I. This conflict immediately derives the badness of sentences whose infelicity we have attributed to the Beaver-Clark constraint, such as (55).

Merchant himself goes on to propose that simultaneous ellipsis of both *only* and its associate is acceptable because the associate in such constructions involves a *secondary* occurrence of focus, which is no longer required to bear actual pitch accent and can now be silenced. But we believe a potentially simpler version of Merchant's approach is available that does not need to distinguish among different flavors of focus and derives the Beaver-Clark constraint from I and II, while remaining compatible with our use of the constraint in this section.

The Hebrew patterns we have been discussing are not limited to constructions in which it is the main verb that raises (with E-extension from its trace failing to silence adverbial 'only'). Much as in English, an auxiliary verb may function as the right-survivor in an ellipsis construction:

(59) ani hayity kone kan bayit (ilu yaxolti), aval yossi lo haya
I would.PST.1MSG buy here house (if could.PST.1MSG), but Yossi NEG would.PST.3MSG
'I would have bought a house here had I been able to, but Yossi wouldn't have.'

If all verb-phrase ellipsis is actually VMV′E triggered by movement of a verbal head, then an elided constituent following an overt auxiliary like *haya* in (59) must have been triggered by the raising of *haya* — followed by extension of the E-feature assigned to the trace in its pre-raised position, leaving *haya* as the right survivor. We then expect any adjuncts above the position in which *haya* is first merged to be excluded from the ellipsis site below it, just as adjuncts of a lexical verb are excluded from VMV′E constructions — while adjuncts below this position should not be excluded.

In (59) above, *haya* was glossed with the English modal *would*, contributing counterfactuality, or X-marking in the sense of von Fintel & Iatridou (2023). In fact, however, *haya* is ambiguous between a counterfactual/X-marking use and a habitual use (roughly corresponding to English 'used to'), as is the case in many languages (Iatridou 2000). We gloss these two uses as *haya*.X and *haya*.HAB, respectively.

Crucially, on both uses, *haya* can right-survive ellipsis of the phrase below it. This is illustrated in (60), where 'genuine' VPE right-survived by *haya* can be used to answer both questions about counterfactual scenarios and about habitual ones.

(60) **Both uses of *haya* can be right-survivors of ellipsis**

Q1: Gil haya šote jain (ilu haya po)? **X-marking**
Gil *haya*.X drink wine (COMP.X was here)
'Would Gil have drunk wine (had he been here)?'

Q2: Gil haya šote jain (ke- še- haya gar birušalaim)? **Habitual**
Gil *haya*.HAB drink wine (when COMP was living in.Jerusalem)
'Did Gil used to drink wine (when he lived in Jerusalem)?'

A: batuax (še-) hu haya
sure (COMP) he *haya*.{X / HAB}
'Surely he {would have / used to}.'

On the theory of ellipsis presented here, both the habitual and the counterfactual readings of the answer in (60) are the result of three steps: (1) *haya* raises to a higher head; (2) its trace is assigned an E-feature; and (3) E-extension copies the trace's E-feature onto the minimal projection containing the trace and its complement.

Interestingly, however, the two uses of *haya* differ when it is the right-survivor in an ellipsis construction whose antecedent contains 'rak' *only* (in both its uses, as an adverbial or as a focus marker on *only*'s associate agreeing with a silent adverbial). As shown in (61)-(62), only on its counterfactual use can *haya* be a right-survivor in such configurations.

Let us continue to assume, as we have in the main text, that silent *only* has the same properties as its overt counterpart, including the requirement in I above that its associate bear pitch accent. Crucially, however, this requirement needs to be satisfied only when *only* is first merged. The satisfaction of this requirement at any point in the derivation indelibly marks the *only*/associate relation as properly established, perhaps because it immediately licenses proper interpretation of the structure. Once the *only*/associate relation has been properly established, it will not matter if some later instance of ellipsis (i.e. E-extension from some higher silenced head) simultaneously suppresses pronunciation of *only* and its pitch-accented associate. On the other hand, if ellipsis occurs earlier in the derivation, after the associate has been merged but before *only* is merged — this is the Beaver-Clark configuration — requirement I above will never be met, and the *only*/associate relation will never be properly licensed. This approach thus incorporates Merchant's proposals in I and II, but crucially without requiring that the generalization in I will be surface-true (i.e. that the constituent marked for pitch-accent at some point in the derivation actually be pronounced when the sentence is uttered). The requirement must merely have been met at some point in the syntactic derivation. This feature of the approach in turn suggests a model in which the syntax communicates with the phonology at regular intervals during the derivation (e.g. at the completion of each maximal projection or phase) — with E-extension overwriting previously assigned phonology with silence.

(61) **Habitual *haya* cannot right-survive ellipsis if *only* is in its antecedent …**

Q: Gil (rak) haya šote (rak) jain (ke- še- gar birušalaim)?
Gil only haya.HAB drink only wine (when COMP live in.Jerusalem)
'Did Gil used to drink only wine (when he lived in Jerusalem)?'

A: #lo batuax (še-) hu haya
NEG sure (COMP-) he *haya*.HAB
Intended but unavailable: 'It is not certain that he used to drink only wine'

(62) **…but X-marking *haya* can**

Q: Gil (rak) haya šote (rak) jain (ilu haya po)?
Gil only haya.X drink only wine (COMP.X was here)
'Would Gil have drunk only wine (had he been here)?'

A:✓ lo batuax (še-) hu haya
NEG sure (COMP-) he *haya*.X
'It is not certain that he would have drunk only wine (had he been here)'

When it is possible to tell, heads that realize counterfactuality/X-marking appear to be structurally higher than heads that realize tense and aspectual categories such as habituality, perhaps associated with C (as argued by Ritter & Wiltschko 2014). In Turkish, for example, as discussed by Aygen (2004) and Bjorkman (2011, 212ff.), when a past tense suffix marks counterfactuality in a conditional, it follows the conditional suffix, but when it marks actual past tense in a non-counterfactual conditional, it precedes that suffix — a sign, given the Mirror Principle, that the counterfactual past-tense head is structurally higher than its real tense-marking twin. The same proposal has been advanced for Hebrew *haya*, which also serves as a past-tense form of the copula, by Boneh & Doron (2008). They argue that both the habitual and counterfactual markers in Hebrew originate as null operators, say HAB and X, that raise to T and get spelled-out as *haya*, which does not itself convey either habituality or counterfactuality, but is rather the morphological realization of tense and agreement on a null operator.[30] If adverbial *rak* must be merged above the position of HAB, but can be merged freely either below or above X, the contrast between (61) and (62) follows immediately.[31]

(63) **Proposed Structure**
(silent-adverbial-*only*) ... *haya*.X ... (silent-adverbial-*only*) ... haya.HAB ... *wine* ...

To achieve the silencing of a phrase below habitual *haya*, *haya*.HAB has to raise, and the E-feature assigned to its trace needs to be extended. But given our formulation of E-extension, E on *haya*.HAB does not extend high enough to dominate silent-adverbial-*only* if that operator is in a position above the pre-raising position of *haya*.HAB. As long as this is stipulated to be the case as in (63), ellipsis triggered by movement of *haya*.HAB will leave adverbial only on one side of an ellipsis boundary and its focus associate on the other. This violates Beaver and Clark's generalization, thus deriving the infelicity of ellipsis under *haya*.HAB in (61) and uniting it with the infelicity of object gaps with *only* in their antecedents under an overt lexical verb observed by Benbaji (2022).

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30. It is the raising of both uses of *haya* to T that makes it difficult or impossible to find word-order evidence concerning their initial positions.

31. Some modifications of the clausal spine proposed by Boneh & Doron (2008) are required for our analysis to proceed. They posit two projections between vP and TP, Aspect and Mood, such that TP $\gg$ MoodP $\gg$ AspectP $\gg$ vP (where $\gg$ indicates dominance). On their analysis, HAB is born as the head of AspectP and raises cyclically (through MoodP) to T, while X is born as the head of MoodP and raises (vacuously) cyclically to T. Our analysis of the infelicity in (61) requires that there be no trace position of HAB between its final landing position and silent adverbial *only*. Had there been such a position, there would have been an E-assigned trace of HAB above adverbial *only* which would have been able to E-extend, predicting (61) to be just as felicitous as (62). However, if movement of HAB proceeds cyclically from AspectP to TP, it passes through MoodP. And if MoodP is the merging position of counterfactual *haya* (X), then it must dominate adverbial *only* for (62) to be felicitous. We can see two ways around this problem. Instead of positing an independent Mood projection for X-marking, we can have the operator X originate in TP to begin with, thus eliminating a potential trace position for HAB. Alternatively, we could claim that the Mood head position is unavailable for HAB as a landing site on its way to T, though it is not clear to us why this should be the case.

This way of ruling out the answer in (61) does not extend to the answer in (62). If adverbial *only* is granted flexibility to merge either above or below the original position of *haya*.X, it can be contained by ellipsis triggered by short-movement of *haya*.X in accordance with Beaver and Clark's constraint.[32]

As Hirsch (2017) points out, all that the semantics of adverbial *only* requires is that it reside above a proposition-denoting constituent for type reasons. Thus, any position above vP can be sensibly assumed to host it. The major stipulation required for our explanation of the Hebrew facts to work is that there is that the maximal projection of *haya*.HAB constitutes a lower bound below which adverbial *only* cannot reside. However, this is a minor departure from the assumptions of Hirsch (2017), as it still allows the operator to be merged into any other verbal domain, T included.[33,34]

8 Restricting the distribution of ellipsis phenomena

In the sections that follow, we will defend a "totalist" proposal concerning constructions traditionally analyzed as ellipsis of verb-headed maximal projections, according to which *all* such constructions actually instantiate verb-stranding maximal V′ ellipsis (VMV′E), including English constructions in which the silenced phrase is headed by an auxiliary verb. If this effort succeeds in the long run, we should not need any independent notion of an external,

32. Note that if in (61)-(62), 'only' can surface not only above the auxiliary or adjacent to the direct object, but in an intermediate position immediately preceding the lexical verb (*drink*) – we would need to make some further non-trivial stipulations to account for this. We judge 'only' in that intermediate position to be somewhat degraded relative to its other potential positions, perhaps not degraded enough to rule it out as completely ungrammatical, but degraded enough to set it aside for the purposes of this paper.

33. No such limitation holds in English, where adverbial *only* behaves as if it may occupy a position below the initial position of any right-surviving auxiliary:

(i) John would only drink WINE when he lived in Cambridge, and Bill would too.
(*may mean* '... and Bill would only drink WINE')

(ii) Mary has been only drinking WINE lately, and Sue has (been) too.
(*may mean* '... and Sue has been only drinking WINE lately too')

(We discuss the analysis of right-surviving non-initial auxiliary verbs, as in the full version of (ii), in section 9.)

34. The interaction of *haya* with *only* illustrated in (61)-(62) constitutes, we think, a challenge for the theory of adjunct exclusion advanced in Landau (2018, 2020a,b, 2022).

In Landau's theory, ellipsis is simply the silencing of a maximal projection. To account for adjunct exclusion in the vP-domain, Landau rules out VP ellipsis after V had raised by positing a system in which movement of a head X that crosses a *spell-out domain* bleeds XP ellipsis. On the (common) assumption that vP and CP are spell-out domain boundaries, raising a lexical verb outside of vP bleeds VPE. As a consequence, any post-raising silencing operation (if available) must target a projection smaller than VP, and must crucially exclude VP-adjuncts from the silenced constituent.

However, AUXP ellipsis is not bled by raising the head AUX, as long as raising does not cross the next-higher spell-out domain; namely, CP. Thus, in particular, AUX-raising to T does not bleed AUXP ellipsis. Recall that in (61) we observe that, like lexical verbs, the Hebrew habitual auxiliary *haya*.HAB seems unable to right-survive ellipsis of a constituent that is presumed to contain *only*. On the other hand, (62) illustrates that counterfactual *haya*.X can right-survive ellipsis of a constituent that contains *only*. If, as assumed by researchers including Boneh & Doron (2008), both the habitual and X-marking auxiliaries raise to T for inflection, their diverging interactions with *only* and ellipsis remain unaccounted for in Landau's system. Given that neither of these auxiliaries crosses a spell-out domain by raising to T, silencing their maximal projections – or, in fact, any maximal projection between their original merging position and T – is not blocked by anything in Landaus' system. Therefore, we would expect both auxiliaries to surface as right-survivors of ellipsis operations that silence constituents containing *only*, rather than the divergence actually attested, where one patterns with lexical verbs and the other does not.

On our approach (in contrast to Landau's), once we accept that what determines the availability and size of an elided constituent is the original pre-movement position of the head of the silenced phrase, rather than the derived position to which that head raises, the unavailability of *haya*.HAB as the right-survivor of silencing of a constituent containing *only* can be reduced to an instance of adjunct exclusion, leading to a violation of the independent Beaver-Clarke constraint on association with focus across ellipsis boundaries (54).

ellipsis-specific licensor of ellipsis of the sort argued for by Lobeck (1995), Aelbrecht (2010) and others — just a higher head that triggers raising of a verb to it, plus the possibility of E-extension from the trace of the raised head.

A totalist proposal of this sort thus predicts that whenever a verb-headed phrase appears to have been elided, its head has raised out of it to some higher position. We will present some arguments supporting this prediction. At the same time, however, we have to note that the converse does not appear to be true. It is not the case that whenever a verb has raised to a higher position, E-extension is guaranteed to be possible, yielding an apparent instance of verb-phrasal ellipsis. English boasts a wide range of verb-projection silencing configurations, which we will be discussing below, but many languages severely restrict such possibilities (or lack them entirely).[35] For example, though French has robust finite V-to-T movement, Authier (2011, 192) shows that the only possible right survivors in verbal ellipsis constructions are the modals *pouvoir* 'be able', *devoir* 'must/should', *vouloir* 'want', *falloir* 'be necessary', *avoir le droit* 'be allowed' — and then only under specific readings. Besides these four modals, no other verb may function as a right survivor. Other modals, the auxiliary verbs *être* 'be' and *avoir* 'have', and main verbs moved to T are all excluded from serving this role.

This is a problem similar to that faced by our analysis of sluicing in section 4, where we noted the impossibility of sluicing in English relative clauses (and counterparts cross-linguistically), even when a null C should in principle permit E-extension. Once again, though our proposals do not provide an explanation for such limitations, they do have a pigeon hole for them (which could ultimately turn out to form part of a deeper explanation): permitting a specific lexical item to stipulate that its trace does not permit E-extension. As was the case with our discussion of limitations on sluicing, we acknowledge that a deeper explanation would be desirable, and we note that this proposal is reminiscent of – but crucially not identical to – the notion of "licenser" for ellipsis that we otherwise avoid here.

Though English permits more instances of E-extension from verbal traces than French does, a similar problem arises in English when the raising of V to *v* is considered. There is strong evidence from the "raising-to-object" constructions that main-verb V obligatorily raises overtly to *v*. For example, in a sentence like (64a), the embedded subject raises into the higher VP, crossing an adverb that belongs to the higher clause (a possibility discussed first by Postal 1974, 146-8) — followed obligatorily by the raising of the higher verb over the raised subject and adverb, which we assume to be movement to *v*. Failure to raise the verb yields ungrammaticality, as (64b) shows:

(64) **Obligatoriness of V-to-*v* raising (English)**

a. Mary **proved** +*v* [$_{\text{VP}}$ Bacon conclusively ___ [___ to have written *Hamlet*]].

b. *Mary *v* [$_{\text{VP}}$ Bacon conclusively **proved** [___ to have written *Hamlet*]].

A parallel argument can be made on the basis of the English verb-particle construction (Johnson 1991). Clearly we need to stipulate that E-extension is blocked from the trace of main-verb V in English.

In Hebrew, by contrast, E-extension from the trace of V must be permitted. Adjunct Exclusion under verb-stranding ellipsis of the type studied above holds of all adjuncts in the verbal domain, which our proposal explains as a consequence of E-extension from the trace of the main verb silencing the minimal V′ but nothing higher. In Hebrew, however, the main verb does end up in T (Shlonsky 1997, 30ff), at least as an option (Borer 1995) — which creates a problem. Though E-extension from V will correctly exclude all adjuncts, low and high, from the silenced domain, if V moves first to *v*. and then V+*v* moves as a unit to T, E-extension from the trace of *v* should make it possible to include low VP-level adjuncts in the silenced domain, providing an end run around Adjunct Exclusion. We have been unable to find any kind of adjunct, low as well as high, that is not excluded from verb-stranding ellipsis in Hebrew. We thus need to permit E-extension from the trace of V, but forbid it from the trace of *v*.[36] If no language with

35. Saab (2022a) presents an interesting argument that those Romance languages such as Spanish that disallow English-style verbal ellipsis constructions also lack verbal head movement — despite appearances to the contrary. What one might take to be head movement of a main or auxiliary verb, he argues, represent a distinct restructuring process.

36. If one assumes a VP-shell analysis of ditransitive VPs, positing a single instance of V whose movement creates multiple head positions within the VP (Larson 1988), we must stipulate that E-extension is only licensed from a trace in the highest of those VP-internal positions.

main-verb stranding verb-projection ellipsis permits an end run of this sort, we may speculate that this restriction of E-extension from *v* is universal, in contrast to E-extension from the trace of V, which languages permit to varying degrees.

Reviewing the languages discussed in this section, we thus make the following provisional proposal concerning restrictions on E-extension from the traces of various sorts of verbs:

(65) **Restrictions on E-extension from verbal traces**
  a. **Universal**
  E on *v* may not be extended.
  b. **Language-particular**
  English: No E-extension from the trace of [-AUX] (i.e. main-verb) V.
  French: No E-extension from the trace of V unless it is *pouvoir* 'be able', *devoir* 'must/should', *vouloir* 'want', or *falloir* 'be necessary' (with specific types of semantics).
  Hebrew: E-extension possible from the trace of any V.

9 Auxiliary verb movement and E-extension

If our proposals are correct, wherever one finds a silenced verb-headed phrase (i.e. apparent VP ellipsis), the right-surviving head must have undergone movement, with E-extension from its trace silencing its original complement. Though we have just seen that E-extension must be blocked from applying to the trace of an English [-AUX] main verb that has raised to *v*, no comparable general restriction appears to hold for [+AUX] verbs.[37] We have already had a glimpse of E-extension applying to the the trace of auxiliary verb *haya* in Hebrew — but our focus in that section was higher-element exclusion phenomena predicted by our proposal, and we did not offer direct arguments that auxiliary verbs do in fact raise. We remedy that in this section, with a focus on English. We will see that E-extension is also possible from the trace of any raised auxiliary verb in English, whether it has raised to T or to a lower head. Additionally and crucially, we will show that when an auxiliary verb does *not* raise to a higher position, it cannot serve as a right-survivor for E-silencing. This observation will thus provide a new argument for the Uniformity of Silence conjecture and the E-extension theory of ellipsis phenomena. In addition, our proposal will motivate a general structure for the English auxiliary system closely aligned with proposals that treat them as raising verbs of a fairly normal sort (Ross 1969a; Pullum & Wilson 1977), and will provide a new perspective on how the morphology governed by auxiliary verbs is assigned.

We begin by noting that when the right-survivor in a verbal ellipsis construction is a finite auxiliary verb *have* or *be* in any of its usages (progressive BE_{prog}, passive BE_{pass}, copular BE_{cop}), as seen in (66), the E-extension theory of such constructions straightforwardly explains its ability to serve in that capacity. Following Emonds (1976, 213) and Pollock (1989), we assume that *have* or *be* has raised to T from a lower position in examples of this sort, responding to a probe on T that seeks the closest bearer of the feature [+AUX]. The trace of the [+AUX] verb that has moved to T will be assigned an E-feature — which, when extended, will silence its complement, yielding an ellipsis construction.

(66) ***Have/be* raising to T plus E-extension yielding a verbal ellipsis construction**
  a. Mary has written a letter, and Sue has too.
  b. Mary was writing a letter, and Sue was too.
  c. This letter was written by Mary, and that one was too.
  d. Mary is a syntactician, and Sue is too.

Comparable ellipsis constructions in which the right-survivor is a modal can be explained in the same fashion, so long as we posit an initial location for modals lower than T, with verb raising applying to it as it does to *have* and *be* in examples like those in (66). This is not a point on which all analyses have agreed (the alternative being the proposal that a modal simply *is* an instantiation of T; cf. Emonds 1976, 213; Lasnik 2000, 161-162 for discussion) —

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37. We will use the feature [AUX] to distinguish main verbs (bearing [-AUX]) from auxiliary verbs and other forms, such as the copula, that pattern with them (bearing [+AUX]). We will assume a feature geometry in which the feature [AUX] is available only to verbs. The features that distinguish modals, auxiliary *have*, and the various uses of *be*, which will be important below, we also assume are subfeatures of [AUX].

though evidence bearing on the issue has always been sparse. If our proposals are correct, they help settle the matter in favor of a lower initial position for modals, followed by raising to T:[38]

(67) **Modal raising to T plus E-extension yielding a verbal ellipsis construction**
a. Mary will write a letter, and Sue will too.
b. Bill can speak French, and we can too.

Auxiliary V-to-*v* raising. All of the ellipsis configurations just discussed, if we are correct about modals, are unsurprising in the context of our proposals in this paper — since movement of the highest auxiliary verb to T is an independently well-supported hypothesis. More surprising, however, is the fact that when more than one [+AUX] verb is present, lower members of the stack may also function as right-survivors in a verbal ellipsis construction, as illustrated in (68) below. The absence from (68) of forms bearing the suffix -*ing* is deliberate. We return to such cases below.

(68) **[+AUX] verbs not raised to T as right-survivors**
Mary will have been invited to the party by then, and...
a. …Sue will have too.
b. …Sue will have been too.
c. Mary will be invited to the party, and Sue will be too.[39]

The proposals advanced here compel us to analyze these examples as a consequence of the right-surviving [+AUX] verb having raised a short distance to a position somewhere below the next highest [+AUX] verb — with the E-feature on its trace extended to the minimal single-bar projection of its original position. The question that needs addressing is the identity of the position to which it moves.[40]

We propose that the "extended projection" (Grimshaw 2005) of every [+AUX] verb may in principle include a position to which that verb may move — with the E-feature of its trace capable in principle of undergoing optional E-extension. In (68a), for example, the complement of *will* is an extended projection of *have*, which includes a position to which *have* raises, with E-extension from its trace yielding, in effect, *have*-stranding *have′* ellipsis. We propose that this position is a little-*v* twinned with big-V *have* just as every other instance of main verb V is arguably twinned with a higher head *v* that selects it — in keeping with proposals that identify auxiliary verbs as otherwise normal instances of V with an additional [+AUX] feature. Raising of auxiliary *have* to its own *v* associates an E-feature with its trace, as always, and optional E-extension from that trace will yield an ellipsis construction. An exactly comparable analysis can be given to (68b) and (68c), with *be* raising to its own *v*.

The kind of structure that we propose for stacks of auxiliary verbs is illustrated in (69). E-extension from the trace of V_{MOD} yields examples like (67);[41] E-extension from the trace of V_{PERF} yields examples like (68a); and E-extension from the trace of V_{PROG} yields examples like (68b) and (68c).

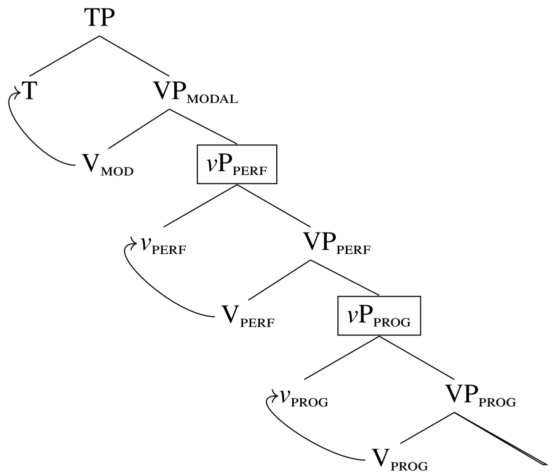
38. The fact that English modals do not occur in nonfinite clauses, sometimes taken as an argument that they are instantiations of finite T themselves, can be understood in a number of different ways. For example, in the spirit of such theories, one might assign them to the same category as modals, but posit that they too originate in a lower position than T and raise to that position.

39. Copular BE_{cop} behaves similarly, as expected; cf. *Mary will have been a professor for a while by then, and Sue will have been too.* and *Mary will be here soon, and Sue will be too.*

40. Our proposal has an ancestor in Thoms (2010), which was an important influence on our work in this section. Thoms proposed a novel structure for the English auxiliary system that raises each AUX to a (much) higher position — and views this movement as a necessary precursor to ellipsis. He suggested that ellipsis is a repair strategy when the trace of head movement has erroneously not been silenced — silencing instead the sister of the head's new position (potentially a much larger domain than we believe can be elided). This is a point on which Thoms' proposal differs crucially from ours, which attributes ellipsis constructions to E-extension.

41. Diagram (69) does not show a *vP* above VP_{MODAL}, though it is possible one should be included. Raising of the modal directly to T is sufficient to account for ellipsis constructions in which the modal is the right-survivor. If the modal were to raise first to a *v* between it and T, and then modal+*v* were to raise to T as a unit, we might predict the possibility that low adjuncts associated with the modal could be included in the ellipsis (i.e. an exception to adjunct exclusion) — unless E-extension for the modal's *v* were blocked, as we must block comparable E-extension for the *v* associated with main verbs (as discussed in section 8). Though we have tried to find evidence bearing on the issue, we have not been successful. If there is indeed no distinct *v* between T and VP_{MODAL}, it would raise the possibility that T is itself the *v* associated with the modal. We will not pursue this question further, however.

(69) **Structure of the upper region of the English auxiliary verb system (preliminary)**
Extended projections marked with boxed node labels for clarity



This proposal is inspired by Harwood (2014), who also proposed that the right-surviving verb in constructions like those of (68) moves a short distance, much as just discussed. Harwood, however, characterized the nature of the landing site in a different manner than our proposals make available to us. It is a famous property of the English auxiliary verb system that it obeys the strict template MODAL>*have*$_{\text{PERF}}$>*be*$_{\text{PROG}}$>*be*$_{\text{PASSIVE}}$>MAIN-VERB — each element of which is optional, but determines the morphology of the verb that comes next (auxiliary or main) *whatever that verb may be* (Chomsky 1957). In *Syntactic Structures*, this logic was captured by first introducing each auxiliary verb next to the affix that it assigns, with a later rule ("Affix Hopping") moving the affix onto the verb to its right. Harwood's proposal follows a similar path: each verb acquires its morphology by raising to a head occupied by the affix that ends up attached to it — where that affixal head is itself selected by the immediately higher auxiliary verb (whatever that verb may be). Each member of the AUX sequence in Harwood's proposal obligatorily subcategorizes for a phrase headed by the affix that the next verb is required to bear (which it acquires by movement).

Our variant of Harwood's proposal does not relate the short movement of these verbs to the morphology it ends up bearing, but to the presence in its extended projection of a head (*v*) with a V-probe that triggers movement. The reasons for our deviation from Harwood's proposal are twofold. First, our proposal permits an instructive account of the special behavior of *be* when it bears the affix -*ing*, to which we turn below — which provides indirect additional support for our theory of ellipsis phenomena, while simultaneously motivating our alternative to Harwood's proposal. In addition, as we shall see, our variant of Harwood's proposal permits a straightforward adaptation of Merchant's (2013) account of voice mismatches between an elided verbal constituent and its antecedent that does not run afoul of our conclusions concerning verbs affixed with *-ing*.

If the structure of the English auxiliary system is as sketched in (69), and does not posit any version of Affix Hopping, what does account for the affixes required by various members of the English auxiliary system? As it happens, the property of singling out the "next X down the tree, whatever it may be", is a familiar property of *probes* in search of an appropriate *goal*, and of agreement phenomena in general. Furthermore, if a probe finds a goal with which it interacts, it is not only morphology on the head with the probe that can be produced by this relation ("agreement") but also morphology on the *goal* — a possibility dubbed "goal flagging" by Deal (to appear). This is the logic behind proposals that permit successful probing by a φ-probe, for example, to yield case morphology on its goal, as a "reflex of agreement" (Chomsky 2001, 16; citing George & Kornfilt 1981) — potentially extended to other syntactic configurations and their morphological consequences.

Building on Deal's ideas, we propose that it is a V-seeking probe on T and auxiliary verbs that is responsible in English for the characteristic morphology that each element deposits on the next verb down the tree (whatever that verb may be) — and that this morphology is simply another instance of goal flagging. The famous "whatever that verb may be" property of the system results from the normal logic of probing, which looks for the closest bearer of the feature set that the probe cares about. For morphology assigned to the highest verbal element by finite T, a similar proposal was in fact already advanced by Pesetsky & Torrego (2007, 270 ff.). Here we extend it to all the affixes assigned by elements in the English verbal auxiliary system. Thus, finite agreement and tense morphology on the highest auxiliary or main verb flags it as the goal found and agreed with by a V-probe on finite T; participial morphology on the highest auxiliary or main verb below *have*$_{\text{PERF}}$ flags it as the goal that *have*$_{\text{PERF}}$'s V-probe found and agreed with; *-ing* morphology on the highest auxiliary or main verb below *be*$_{\text{PROG}}$ flags it as the goal that *be*$_{\text{PROG}}$'s V-probe found and agreed with; and participial morphology on a passive verb flags it as the goal found and agreed with by *be*$_{\text{PASS}}$.

Crucially, this proposal makes a sharp distinction between answers to the question "what does T or AUX assign morphology to?" and "what category does T or AUX subcategorize for" — since the answer to the first question will be determined by probe-goal locality, which can span distances greater than the sisterhood relation that restricts subcategorization (Chomsky 1965, 99). The special behavior of verbs affixed with *-ing*, to which we now turn, provides an argument in favor of this distinction, while demonstrating the important correlation between head movement and ellipsis phenomenon predicted by our overall proposal.

The special status of *-ing*. Sag (1976, 27) and Johnson (2001), among others, have noted an exception to the possibilities documented above in (68). A [+AUX] verb affixed with *-ing* may not serve as a right-survivor in an ellipsis construction. Because main verbs are independently impossible as right-survivors in English, due to the absence of E-extension from V (as discussed in section 8), relevant cases always involve *be*$_{\text{PASS}}$ or *be*$_{\text{COP}}$ embedded under *be*$_{\text{PROG}}$:

(70) ***Verb+*ing* as right-survivor of ellipsis**
   a. *Doc Golightly is being discussed and Sally is being too. (Johnson 2001, ex. 12)
   b. *Mary might have been being arrested then, and Sally might have been being too.
   c. *John is being obtuse, and you are being too.

From a morphological standpoint, nothing special needs to be said about the suffixation of *-ing* on *be* in such cases. The suffix *-ing* can be viewed as a goal flag that appears on any verb successfully probed by BE$_{prog}$, as just discussed. What must be excluded is the possibility of raising *be*$_{\text{PASS}}$ to *v* below *be*$_{\text{PROG}}$. Severing the subcategorization properties of auxiliary verbs from the morphology that they apply to the next verb down makes this possible. We propose, once again inspired by observations of Harwood's (2014), that BE$_{prog}$ has the peculiarity that when it merges with an extended projection headed by a second instance of *be*, it requires that phrase to be a bare VP rather than a full *vP*. As a consequence the complement to BE$_{prog}$ will contain a verb that bears *-ing* morphology (goal-flagging agreement with a V-probe on BE$_{prog}$), but that verb cannot be the right-survivor in an ellipsis configuration, because it does not raise.[42]

Our claim that [+AUX] verbs goal-flagged by BE$_{prog}$ do not raise is of course a stipulation — but it is a stipulation that receives independent support from constructions unrelated to ellipsis, studied by Harwood (2014, 311-314). Harwood examined English existential *there*-constructions in clauses with multiple auxiliary verbs. These constructions all contain some form of *be*, and an "associate" nominal. He noted a striking asymmetry in the relative positioning of the associate nominal and forms of *be* with and without morphology assigned by BE$_{prog}$. In particular, the associate nominal must follow an instance of *be* that has not been suffixed with progressive *-ing* — but if *be* does bear progressive *-ing*, the associate precedes it instead.

This is not a novel empirical observation. Since a form of *be* suffixed with progressive *-ing* will have received this morphology thanks to the presence of BE$_{prog}$ to its left, it amounts to the "leftmost *be* condition" of Milsark (1974, 65), which requires the associate to follow the leftmost occurrence of *be* in the string of auxiliaries. If taken at face value, however, a "leftmost *be* condition" is peculiar, since it posits a nonlocal selectional property that is unique to this construction. Harwood's important contribution is the observation that one can explain the facts behind this condition in non-*sui generis* terms if we suppose that (A) the associate is merged as a specifier of *be* (a very normal instance of selection); and (B) the order *be* > associate results from obligatory raising of *be* over its specifier to a higher head (another normal process) — but with an exception: (C) an auxiliary verb that heads the complement

42. BE$_{prog}$ appears to have no problem merging with a full *v*P extended projection headed by a main verb — so we cannot generalize its aversion to *v*P complementation to include main-verb *v*P. Consider, for example, the availability of Raising-to-Object constructions in clauses with progressive *be* such as: *Mary has been believing Sue quite sincerely to be the best candidate*. If such constructions involve raising of the embedded subject to spec,VP of the higher clause over main-clause VP adverbials; (Postal 1974, 146-147), followed by V-to-*v* raising (which positions the higher verb to the left of the raised subject), it is clear that progressive *be* may take a full *v*P as its complement so long as the head of the extended projection *v*P is a main verb.

to BE$_{prog}$ does not undergo raising. Exception C is of course exactly what our proposal needs to state to explain the inability of such a form to function as the right-survivor in an ellipsis construction — so Harwood's observation constitutes indirect independent evidence for our proposal.[43]

(71) **Raising of *be* over the associate in a *there*-construction (adapted from Harwood 2014, 311-314)**

Raising required:

a. There will **be** many smurfs — arrested for anti-social behaviour.
b. *There will many smurfs **be** arrested for anti-social behaviour.

c. There have **been** many smurfs — arrested for anti-social behaviour.
d. *There have many smurfs **been** arrested for anti-social behaviour.

e. There must **be** many smurfs — dancing in the fields.
f. *There must many smurfs **be** dancing in the fields.

g. There have **been** many smurfs — dancing in the fields
h. *There have many smurfs **been** dancing in the fields

No raising:

i. *There were **being** many smurfs — arrested for anti-social behaviour.
j. There were many smurfs **being** arrested for anti-social behaviour.

A preliminary sketch of our proposed structure for the English auxiliary system, including subcategorization properties, probes with their goal flags, and arrows indicating movement to T and *v* is shown in (72) below. The maximal extended projection of each verb is indicated by a boxed node. A few notes about the choices we have made:

Features: We assume, as stated in note 37 above, that the feature AUX is defined only for verbal elements, so our subcategorization statements do not need to mention verbal features if they include a value for [AUX]. Similarly, the features that distinguish among the various auxiliary verbs are subfeatures of [+AUX], so we have not specified that feature either when [MOD], etc. are mentioned in subcategorization statements. We assume that some property distinguishes *v*P from VP, but have chosen to just write their names when needed (for the subcategorization of *be*$_{\text{PROG}}$). Main verbs are distinguishable as [-AUX].

Subcategorization: We use subcategorization statements to capture the rigid order in which the various auxiliaries are permitted to stack, though of course there might be deeper explanations.[44] The dual subcategorization specification for BE$_{prog}$ reflects the observation in note 42 that main verb V has a normal *v*-V extended projection even when it is selected by BE$_{prog}$. We assume that the [AUX] value of V is shared by its *v*, perhaps as a consequence of agreement, and that features of heads are also properties of their maximal projections.

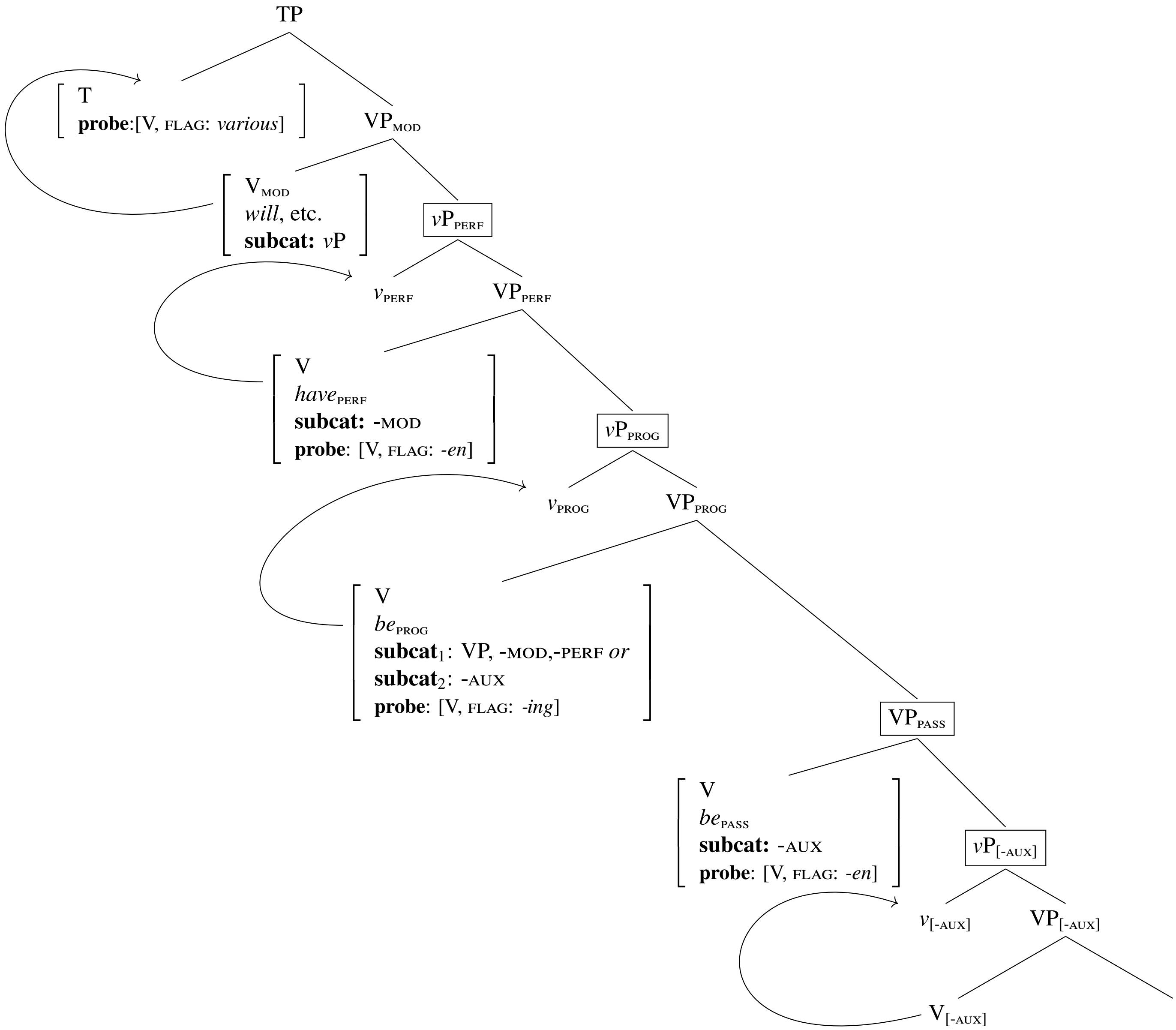
Probes and Goal flagging: Each V-probe is satisfied by the closest verb (whether [+AUX] or [-AUX]) that interacts with it, and assigns the indicated flag as morphology to it. The tree incorporates the proposal that a modal has no

43. Harwood (pp. 329ff.) ultimately calls this conclusion into question and rejects it, wrongly we think, because of adverbs that he believes should precede *being* if there were no raising of that auxiliary, but instead prefer to follow it. To the extent that we agree with his claims about the facts, we believe that the adverbs at issue might simply be main-verb modifiers, following the auxiliary because they were merged lower in the first place.

44. To reduce clutter, we have not included additional detail in our subcategorization statements that would prevent multiple occurrences of the same auxiliary, such as multiple modals, multiple perfect *have*, etc. We can do this by adding [-MOD] to the subcategorization statement for modals, and similarly for the other auxiliaries, but leave it open whether this is the correct approach, or whether some other factor blocks multiple occurrences of auxiliaries. As noted on the "Multiple Modals" page of the Yale Grammatical Diversity Project site (Huang 2011), dialects that permit multiple modals move either the second modal over the first or both together in T-to-C questions, which suggests a poverty-of-the-stimulus argument that something deeper than arbitrary subcategorization blocks structures in which a modal embeds a second modal-head phrase — an analysis that otherwise might have been the most obvious analysis of multiple-modal constructions (or at least a plausible one) for the child acquiring a dialect that permits multiple modals.

probe that produces goal flagging, since the verb that such a probe would find as its goal appears in its bare uninflected form. Alternatively, it could be represented as assigning a null affix as its flag. We know of no evidence bearing on this choice, and have picked the more parsimonious alternative.

(72) **Structure of the English auxiliary verb system**



Implications for voice and voice mismatches. The proposals just sketched have implications for the syntax of voice (at least in English) and the phenomenon of "voice mismatch" in ellipsis explored by Merchant (2013) — especially our claim that the morphology assigned by T and the members of the auxiliary verb system instantiates goal flagging in the sense of Deal (to appear). This subsection outlines how Merchant's observations and proposal can be easily adapted into the overall picture that we have painted of the English auxiliary system, and what they tell us about how syntactic identity conditions interact with the view of verb-projection ellipsis that we have developed so far.

The structure illustrated in (72) is maximal, containing an instance of every category of auxiliary verb. English does, of course, also permit one or more auxiliary verbs to be missing. We assume that when an auxiliary verb is missing from a sentence (for reasons other than ellipsis), it has never been merged in the first place. The subcategorization properties the auxiliary verbs listed in (72) make this possible. For example, in a derivation in which *be*$_{\text{PROG}}$ is not merged (but all the other heads in (72) are), the [-MOD] subcategorization property of *have*$_{\text{PERF}}$ would be satisfied just as well by a complement headed by *be*$_{\text{PASS}}$ as it is by a *be*$_{\text{PROG}}$-headed complement in (72).

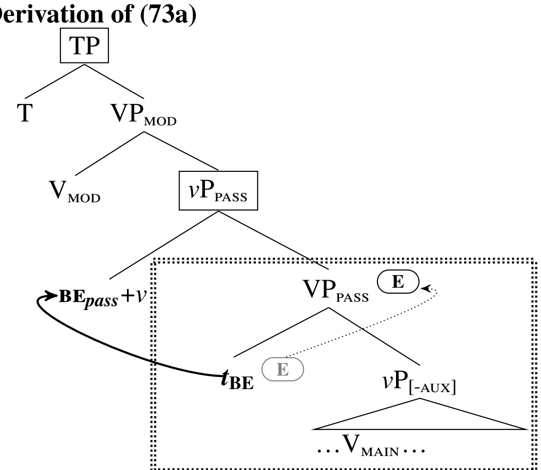
Recall now that it is a property of *be*$_{\text{PROG}}$, not a property of *be*$_{\text{PASS}}$, that is responsible for the absence of a *v*P projection above *be*$_{\text{PASS}}$, the factor that explains the inability of passive *be-ing* to function as a right survivor in an ellipsis construction. If *be*$_{\text{PROG}}$ is never merged in the first place, *be*$_{\text{PASS}}$ will have its own *v* to raise to — and if it is the only auxiliary verb present, it will raise to T. E-extension from its trace will yield ellipsis configurations like (73a-b):

(73) **E-extension from raised *be*$_{\text{PASS}}$**

a. ***be*$_{\text{PASS}}$ raised to *v***
Mary will be invited to the party, and Sue will be too. *repeated from* (*68c*)

b. ***be*$_{\text{PASS}}$ raised to T**
This letter was written by Mary, and that one was too. *repeated from* (*66c*)

(74) **Derivation of (73a)**



In the introduction to this paper, we noted that we would be presupposing a distinction between "(i) the laws of syntax and phonological interpretation responsible for the existence and distribution of ellipsis configurations and (ii) the laws that govern the interpretation of silenced material" — and that we would focus almost all our attention on the former, while assuming at a minimum that a syntactic identity requirement forms part of the latter. Let us view (ii) as forming part of the procedure used by a hearer of an utterance that contains silenced material, faced with the task of "reverse engineering" the derivation of that material as intended by the speaker, following Pesetsky (2021).

A speaker confronted with an utterance containing a clause whose derivation contains the ingredients of (74) will be able to account for the silenced trace of *be*$_{\text{PASS}}$ without reference to syntactic identity with a parallel phrase elsewhere in the discourse, since this is a VMV′E configuration, and the head of the trace's chain remains audible. Turning now to the *v*P complement of *be*$_{\text{PROG}}$, the same will be true for the trace of any argument that has undergone internal or external merge as the specifier of this *v*P and then raised to a higher position (such as the specifier of TP). It is material silenced by E-extension that is neither intrinsically silent like *v* itself nor assigned E because it is a trace that poses a problem for the hearer — a problem solvable, we suggest, only by copying from a syntactically identical antecedent. In general, in constructions like these, with the trace of *be*$_{\text{PROG}}$ and the head and specifier of *v*P already accounted for, the elements that will need to rely on a syntactically identical antecedent are the main verb (raised to *v*) and its syntactic dependents.

If the morphology that is assigned to verbs by T and by other verbs throughout the auxiliary system is goal flagging, it is not a surprise that it is generally ignored as irrelevant to calculations of syntactic identity (Sag (1976, 16ff.)), as is well-known:[45]

45. The morphology of finite *be* appears to be exceptional in this respect, failing to count as syntactically identical to participial and bare counterparts, as discussed by Warner (1986) and Lasnik (2000, 195-6) (though Warner notes that the English of earlier centuries appeared to lack this restriction). We leave this issue and a few similar anomalies unexplained here.

(75) **Goal-flagging mismatches tolerated by ellipsis**

a. Sue is finishing her homework now, but Mary already has.

b. John plays the violin, but Bill can't.

It should thus be no surprise, given structures like (74) that the difference between a verb phrase headed by a main verb with and without the participial morphology assigned as a goal-flag by the V-probe on *be*$_{\text{PASS}}$ is ignored when calculating syntactic identity. This permits active-passive mismatches in both directions, a phenomenon whose existence and properties were documented and studied in depth by Merchant (2013), whose findings we rely on here.

(76) **Voice mismatches in "VP-ellipsis"**

a. **Active→Passive**
The janitor must remove the trash whenever it is apparent that it should be $\triangle_{passive}$.

b. **Passive→Active**
'Slippery slope' arguments can be framed by consequentialists (though I wouldn't $\triangle_{active}$ in this case).
(Merchant 2013, 78-80)

As Merchant demonstrates, the possibility of voice mismatch in ellipsis is limited to situations in which syntactic identity is obeyed, with main-verb morphology the only admissible difference between antecedent and ellipsis site. Crucially, the silenced domain must be a portion of structure that is neutral between active and passive, and the right theory of voice must countenance the existence of such a domain. The minute a larger structural domain is silenced that includes material (besides main-verb morphology) that distinguishes the voices, mismatch is impossible. Merchant demonstrates this with sluicing (which we have analyzed as E-extension from C), which fails to permit voice mismatch:

(77) **Voice mismatches excluded in sluicing**

a. **Active→Passive**
*Someone murdered Joe, but we don't know by whom $\triangle_{passive}$.

b. **Passive→Active**
*Joe was murdered, but we don't know who $\triangle_{active}$. (Merchant 2013, 81, ex. (5))

Our account must differ from Merchant's in one respect, however. Merchant posits a silent Voice head distinct from auxiliary *be*$_{\text{PASS}}$ as the element that determines whether the clause will have active or passive syntax. If we were to follow his lead and posit a silent voice head below *be*$_{\text{PASS}}$, we might expect it to license E-extension on its own — or to raise to its own *v*, with its trace licensing E-extension. This would yield structures in which *be*$_{\text{PASS}}$ would *sound* like the right-survivor, but the actual right-survivor would be Voice. If *be*$_{\text{PASS}}$ were embedded under *be*$_{\text{PROG}}$, this possibility would wrongly provide an end-run around the generalization discussed in the previous subsection — where we blocked passive *being* as a right-survivor by requiring it to lack a *v*P when embedded under *be*$_{\text{PROG}}$. If a homophonous ellipsis construction could be created by E-extension from a distinct (possibly moved) Voice head, we would have a way of deriving forms that are actually blocked. Consequently, we conclude that the element that renders an otherwise active clause passive is actually *be*$_{\text{PASS}}$ itself. In other words, it is the passive auxiliary *be* that functions as the voice-determining head in an English passive clause, and it is its complement *v*P that is indifferent between passive and active syntax.[46]

The syntax of expletive *do*. We have so far ignored the syntax of semantically vacuous (expletive) *do*, famous for being obligatory in a variety of environments, and forbidden in others — which interacts instructively with VP ellipsis constructions. We turn to expletive *do* in this subsection.

Many approaches incorporate (in different ways) a perspective inherited from Chomsky (1957), according to which expletive *do* is introduced whenever finite T finds itself in a position too distant from any verb to assign its morphology to it — and is never introduced otherwise. One of the reasons this characterization has remained attractive is the apparent heterogeneity of the environments that mandate *do* on standard analyses, with the absence of other auxiliary verbs and T's separation from the main verb the only obvious thread uniting them:

46. We have obviously not formulated a full theory of passive syntax here, especially the difference between realization of the external argument as a normal DP in an active sentence and its realization as an implicit argment *by*-phrase in the passive. We do note that the very fact that *be*$_{\text{PASS}}$ licenses a null realization of the external argument below it relieves the hearer of the burden of reverse-engineering its structure in such cases on the basis of a syntactic antecedent.

(78) **Environments of expletive *do***

a. **Auxiliary verb raising to C**

i.e. clauses with T-to-C movement in which T has moved on its own, due to the absence of any [+AUX] modal, *have* or *be* that might have raised to T and then traveled with it to C

Did Mary solve the problem?

b. **Presence of polarity head**

i.e. clauses with polarity element (negation or contrastive affirmation) intervening between T and the nearest verb, due to the absence of any occurrence [+AUX] modal, *have* or *be* that might have raised to T over the intervener[47]

Mary did not solve the problem.
Mary did *so* solve the problem.

c. **VP-ellipsis constructions**

i.e. VP ellipsis constructions in clauses lacking any [+AUX] modal, *have* or *be* that might have raised to T serve as a right-survivor

Sue solved the problem and Mary did too.

It is often suggested that *do* is merged directly with T under these circumstances, as a "last resort" remedy that permits finite T to assign its morphology to this newly inserted verbal host. Under the proposals advanced here, however, *do* cannot be analyzed as making its first appearance that high in the structure, if we are to explain environment (78c) (i.e. VP ellipsis constructions). If such constructions are always a consequence of E-extension from the trace of the right-surviving verb, then *do* must have raised to T from a lower position whenever it is the right-survivor in such a construction, just like other [+AUX] verbs that play this role in examples like (66). If *do* raises to T, rather than being born there, in ellipsis constructions, then the use of *do* in environment (78b) — appearing to the left of polarity words like *not* — must be attributed to the same reason other auxiliary verbs end up there: movement to T. If we now add environment (78a) (T-to-C movement) to the discussion, a new thread uniting all three environments suggests itself. In every case where expletive *do* is possible, it reached its final position by *movement*.

Before considering the nature of the movement requirement in greater depth, we must first establish the original position of expletive *do* — the position from which it moves. To this end, we first discuss the category membership and subcategorization property of expletive *do*.

In North American English, expletive *do* never co-occurs with other [+AUX] verbs. This suggests (in the context of our previous proposals) that despite being semantically vacuous, it shares with modal auxiliaries the feature [+MOD]. This feature will prevent it from heading the complement of any other auxiliary verb, since their subcategorization specifications (as shown in (72)) prevent them from merging with phrases headed by [+MOD].[48]

In British English (and some other non-North American dialects), however, expletive *do* does not have this limitation, and therefore may be considered to bear — not [+MOD] as in North American English — but just [+AUX] without further specification. In British English, as discussed by Lewis (to appear) and others whose work she reviews, expletive *do* may be the right-survivor in an ellipsis construction below a modal or $have_{\text{PERF}}$.[49]

(79) **Expletive *do* as right-survivor below modal and $have_{\text{PERF}}$ …**

a. Tom should write a paper and Emma should do too.
b. Tom has written a paper and Emma has done too.
c. Tom should have written a paper and Emma should have done too.

Expletive *do* as a right-survivor is excluded, however, below be_{PASS}, as Lewis notes:

47. In examples with stressed *do*, such as *Mary DÍD solve the problem.* we will assume, following Chomsky (1957, 65), that an "Affirmation" morpheme occurs in the same position as *not* and stressed SO, whose phonology is suprasegmental, placing stress on the element that precedes it.

48. A possible argument against identifying expletive *do* as [+MOD] is the fact that it inflects for third person singular (*does* vs. *do*, unlike other bearers of [+MOD]. We leave this issue unresolved.

49. We owe our understanding of the generalizations concerning the distribution of British English expletive discussed here *do*, as well as the data presented, to Lewis (to appear).

(80) **…but not below *be*$_{\text{PASS}}$ …**
*The pasta has been eaten and the fish has been done too.

Lewis shows that the exclusion of passive clauses as hosts for expletive *be* is not due to any ban on A-movement of the subject from the ellipsis site, since active unaccusative and raising constructions pose no problem for right-surviving *do* (e.g. *The students have seemed to enjoy the class and the professors have done too*). In keeping with our other proposals, therefore, we may attribute the impossibility of examples like (80) in British English to a subcategorization property for *do* that is identical to the subcategorization property of *be*$_{\text{PASS}}$: subcategorization for [-AUX], i.e. phrases headed by main verbs only. The same property will prevent *do* from occurring above *have* and *be* in environments (78a) and (78b):

(81) **Expletive *do* subcategorizes for [-AUX] only**
a. *Did Mary have written the letter?
b. *Did Mary be writing the letter?
c. *Did the letter be written by Mary?

d. *Mary did not have written the letter.
e. *Mary did not be writing the letter.
f. *The letter did not be written by Mary.

We may thus view expletive *do* in both sets of dialects as a semantically vacuous auxiliary verb, optionally merged like any other auxiliary. Like all the other [+AUX] elements of English, its distribution is restricted by its own feature composition and by its subcategorization requirements. We assume the following partial lexical entries for expletive *do*, which are different in detail from the entries for the other auxiliaries, but not different in kind:

(82) **Lexical entry for expletive *do***
a. **North American English:**
category: MOD
subcat: -AUX
b. **British English:**
category: +AUX
subcat: -AUX

Finally, like any other English auxiliary (or main) verb, we propose that expletive *do* may be twinned with a little-*v*, and that it is movement to its *v* followed by E-extension from its trace that produces the ellipsis constructions in (79) in British English. Our hypothesis that expletive *do* is fundamentally just another auxiliary now makes a clear prediction concerning the possibility of expletive *do* below *be*$_{\text{PROG}}$. We already proposed that *be*$_{\text{PROG}}$, unique among the auxiliaries, requires that its complement be a VP and not a *v*P, preventing the verb below it (goal-flagged with *-ing*) from serving as a right-survivor in an ellipsis construction — since it has no *v* to move to. The same should be the case if that verb is expletive *do*. As noted by Lewis, citing Thoms (2011), the prediction is confirmed:

(83) **Expletive *do* never a right-survivor below *be*$_{\text{PROG}}$**
*Rab is throwing a TV out the window, and Morag is doing too. (Thoms 2011, 7 ex. (21b))

We now return to the movement requirement — the salient respect in which expletive *do* is a bit more than "just another auxiliary verb". In fact, our earlier statement of the movement generalization was incomplete, there are derivations in which expletive *do* is introduced in conformity with the lexical entries in (82) and subsequently moves, but still yields an ungrammatical result.

Consider, for example, the much-discussed restriction illustrated in (84a). An unstressed expletive *do* in mainstream dialects of both North American and British English is unacceptable when it ends up directly preceding the main verb without a polarity element such as *not* intervening — even though, all things being equal, there should be a parse in which it has raised to T even if nothing intervenes. A similar issue arises with the British English expletive *do* seen in ellipsis constructions like (79). There expletive *do* raised to *v*, with E-extension from its trace silencing its complement. E-extension is optional elsewhere, however, so *do* should be be able to raise to its *v* even in the absence of subsequent E-extension. Nonetheless the result is unacceptable, as illustrated in (84b):

(84) **Expletive *do* impossible despite having moved**

a. *Mary [$_{T'}$ did [$_{VP}$ — write the letter]]. (*do* to T)

b. *Tom should write a paper and Emma should [$_{vP}$ do [$_{VP}$ — [write a paper too]]]. [British English] (*do* to *v*)

We propose that the key property distinguishing these unacceptable uses of expletive *do* from acceptable counterparts such as those in (78) and British English (79) concerns their interaction with the phonology of linearization. When expletive *do* raises to C over the subject as it does in (78a), or raises to T over a polarity item like *not* as it does in (78b), it precedes material that it would not have preceded if movement had not taken place. Movement of *do* has made a difference to the phonological output. This is not true in the unacceptable (84a), where the movement of *do* is string-vacuous. We suggest that this difference is important: expletive *do*, once introduced into a syntactic derivation, must undergo some operation that has an effect on the phonological output. Examples like (84a) are ruled out because no such operation has taken place.

Now consider ellipsis constructions like (78b), in which expletive *do* has moved to T and is the right-survivor. It is true that expletive *do* here has not moved over an overt polarity item such as *not*, so one might be tempted to describe its movement here too as string-vacuous, and wonder why such constructions are premitted. But unlike expletive *do* in non-ellipsis constructions like (84a), its movement to T does have an effect on its linearization — thanks to E-extension, which silences material to its right that would otherwise have been pronounced, had *do* not raised to T. As a result, once again, expletive *do* fails to precede the same material after movement as it would have preceded had movement not occurred.

Exactly the same account can be given of the behavior of British English expletive *do* when it is merged below other auxiliary verbs. If it moves to *v* but E-extension does not apply to its trace, as is the case in (84a), the movement has no effect on linearization: *do* does not end up preceding material different from what it would have preceded if movement had not taken place. If E-extension does take place, however, the result is once again a derivation in which the raising of expletive *do* made a difference to its linearization.

These observations suggest that movement, the assignment of E to the trace position, and optional E-extension count as a single operation in the syntax, which permits us to state the special property of expletive *do* parsimoniously as follows:

(85) **Special property of expletive *do***
The expletive *do* must undergo an operation that affects its linear order.

Several deep issues are raised by this proposal, of course. One is the fact that it requires the system to calculate and revise linearizations in the course of the derivation, a proposal whose interaction with proposals like Fox and Pesetsky's (2005) "Cyclic Linearization" requires further study. We will leave these questions for future work. Another is the reason a condition like (85) should exist in the first place. The best possible theory would be one in which the properties of expletive *do* — notorious as an example of language-specific idiosyncrasy — actually turn out to result from nothing but universal principles interacting with banal featural and subcategorization properties (which we know differ across languages). Our proposal advances towards that goal, but does not achieve it, because it must stipulate (85). We speculate that the very fact that *do* is semantically vacuous might lie behind (85). Perhaps an element that gives no sign of its presence by creating a change in the meaning of the constituents that contain it must give sign of its presence phonologically, and perhaps a linearization alternation is how that is done. We leave this topic as well for future consideration.

There are a number of more specific loose ends still outstanding, which will raise some issues that will remain unresolved here.

First, though our proposals so far predict the acceptability of expletive *do* in negative sentences like (86a), they do not so far rule out alternatives like (86b).

(86) **Negation requiring expletive *do***
a. Mary did not solve the problem.
b. *Mary not solved the problem.

We believe that this contrast might reveal nothing more than a subcategorization specification of *not* and other polarity heads that rejects main verb complementation. We thus assume that these polarity items subcategorize for [+AUX].

The fact that polarity items subcategorize at all entails that they head projections, as proposed by Pollock (1989) and many others since. One might worry, in keeping with much past discussion, that the structural intervention of a polarity phrase such as NegP between T and the verbal domain might create problems for probing of verbal elements by T. Indeed just this consideration has often been taken to mandate the insertion of expletive *do* in T as a "rescue strategy" for a T that needs to assign morphology to a verb but cannot do so across the polarity phrase boundary. This proposal in turn made puzzling the ability of auxiliary verbs to raise across this boundary, leading to many proposals about how to sidestep or resolve this paradox.

Our approach removes the need to posit any kind of barrier to probing posed by an intervening polarity phrase, and eliminates the need to view the use of *do* in (86a) as any kind of rescue strategy. All we need to assume is that *not* and other polarity heads that occupy the same position do not bear verbal features, and therefore do not establish any barrier to probing by T. As a consequence, we may assume that there is a [+AUX] probe on T that triggers head-movement of the closest bearer of that feature (which may be expletive *do*), disregarding intervening polarity heads just as all probing disregards intervening elements that do not serve as goals. Likewise, we may assume a more general V-probe on T that goal-flags the closest verbal element, again disregarding a intervening polarity head because it is nonverbal. A polarity phrase is not expected to present any barrier to probing by features of T, and does not, on this analysis, thus eliminating a persistent puzzle for the analysis of the English auxiliary system and similar paradigms in other languages (cf. Pollock's analysis of French verb raising).

This approach also casts doubt on the existence of the widely assumed Head Movement Constraint first proposed by Travis (1984, 131). We speculate that a head that blocks the raising of a lower head in response to a higher probe does so not because all heads block head movement (as the Head Movement Constraint claims), but because the specific intervening head in question *satisfies* the probe (adopting the terminology of Deal 2015) and halts the probing process for that reason.

Another loose end concerns a contrast in the logic behind the obligatoriness of raising to T and the obligatoriness of raising to C [+AUX]. We assume that the [+AUX] probe on T comes with the movement-triggering property commonly called EPP. We leave open why it is head movement that is triggered, rather than phrasal movement. This probe appears to follow the logic advocated by Preminger (2011, 2014). When there is a [+AUX] verb present that can raise to T, it must do do. This excludes examples like **Mary not has solved the problem*. But when the structure contains a main verb but no [+AUX] verb, no problem results. There is no verb movement to T (diagnosable by the preverbal rather than postverbal position of low adverbs), but this causes no problem. The logic is that of an obligatory rule that simply fails to apply when its environment is not met.

T-to-C movement in environments that mandate it such as *wh*-questions appears to follow a different logic that we might call "strong obligatoriness". Once again, when a [+AUX] verb is present (whether expletive *do* or some other auxiliary), the highest such element must raise to C, presumably in response to a probe on C with the relevant EPP property. If the structure lacks a [+AUX] verb and contains only a main verb, the result is ungrammaticality — contrasting with the behavior of probing by T just discussed:

(87) **Strong obligatoriness of raising to C**
  a. Which problem did Mary solve?
  b. *Which problem Mary solved?

If expletive *do* is just another optionally chosen auxiliary verb, not added as any kind of "rescue strategy", strong obligatoriness of this sort presents a challenge. Possibly, however, we are not actually observing strong obligatoriness of raising to C in (87b) but something else: subcategorization by this kind of C for a TP whose head's probes have been fully satisfied. If this information is accessible and can be referred to by subcategorization statements, the right distinction can be made between examples like (87a) and (87b), since only in the latter case does T bear an unsatisfied [+AUX] probe with an unsatisfied EPP property. The idea that a C triggering T-to-C raising subcategorizes specifically for TP is supported perhaps by the observation that material cannot be fronted to a position between C and the subject in such clauses that may be placed there in clauses that do not involve T-to-C movement:

(88) **Possible evidence that C triggering T-to-C subcategorizes for TP**
a. I wonder which problem under these circumstances Mary will decide to solve.
b. *Which problem will under these circumstances Mary solve.
c. Tell me how many speakers this year the organizers are inviting.
d. *How many speakers are this year the organizers inviting.

If the material that successfully precedes the subject in (88a) and (88c) has moved into a Topic or Focus projection located between CP and TP, one might take the badness of (88b) and (88d) to result from the failure of C to merge directly with TP. This in turn might provide support for our conjecture that the unacceptability of examples like (87b) might reflect a subcategorization failure on the part of C. We leave this issue too for future work.

10 Conclusions

In this paper, we have argued that a common source underlies circumstances in which an entire phrase is silenced in constructions usually described with the term "ellipsis". We have examined two such configurations in detail, sluicing and verb-projection ellipsis, and argued that in each case the silencing of an entire phrase starts with the silencing of a single lexical item: the complementizer in the case of sluicing, and the trace of verb-movement in the case of verb-projection ellipsis. In each case, we offered independent evidence for the proposal, and showed how the proposal makes additional predictions that appear to be correct. In the case of verb-projection ellipsis, our proposal might seem a bit shocking since it entails that verb-stranding verb-projection ellipsis — which has at best been a side-show in the wider world of VP ellipsis debates and whose very existence has been doubted by some — is actually the right analysis of *all* VP-ellipsis phenomena. We argued, however, that this proposal interacts in illuminating ways with broader discussions about the nature of the verbal system (at least in English), permitting simplifications and advances in our understanding of a variety of puzzles.

Many issues remain uninvestigated, and of course any one of them may prove fatal to the proposals advanced here. Among the more salient unexamined issues are the distribution of null and overt complementizers across languages — crucial to our account of sluicing, but not necessarily correlating as expected with the distribution of sluicing itself; and the wider variety of ellipsis phenomena found in the languages of the world, including the possible existence of argument ellipsis. We leave the discussion of these issues for future work.

Appendix: Notes on Landau's arguments against head-stranding ellipsis

In a series of papers that inspired our research, Landau has argued for an analysis of adjunct exclusion that is very different from ours — building on Hebrew data like (38)-(39) and comparable observations in other languages. He concluded from his findings that these constructions do not instantiate verb-stranding verb-projection ellipsis at all. Instead, the silenced objects in these constructions are derived by *Argument Ellipsis* (AE), which directly suppresses the pronunciation of subcategorized arguments. Since adjuncts are not subcategorized arguments, they cannot undergo AE, and no independent process exists that can silence them, according to Landau's proposal.

Landau compares his approach favorably to previously proposed analyses that analyze these constructions as instances of verb-stranding ellipsis of the entire remnant VP — analyses under which adjunct exclusion is unexpected, requiring some additional stipulation to block the inclusion of adjuncts in the elided domain. Landau proposes to exclude this possibility entirely, positing a universal constraint that blocks verb-stranding VP ellipsis for principled reasons.

Our proposals differ from the verb-stranding ellipsis alternative that Landau considers in positing silencing of not the entire VP, but only the minimal projection in which the verb's subcategorization requirements are satisfied — thus excluding VP-adjuncts. We have offered an array of other arguments for our approach that do not follow from analyses that posit only AE and ban verb-stranding verb-projection ellipsis — in particular, the generalization of adjunct exclusion to constructions with auxiliary verbs and the exclusion of all elements higher than the minimal V′ containing the right-survivor (not just VP adjuncts). At the same time, Landau's work does report a substantial array of constraints on the interpretation of the silenced arguments that, if they have been analyzed correctly, make sense as properties of AE but not of verb-stranding verb-projection ellipsis; at least not straightforwardly. Such findings present an obvious challenge for our proposals insofar as they follow naturally from an "exclusively AE" analysis of silenced objects but not from the expected properties of the silenced verbal projections that we have posited.

In this vein, Landau argues that verbal arguments whose semantic type is not e (i.e. arguments with functional semantic types) resist silencing when the verb remains overt, just the kind of property we expect from a special ellipsis rule that specifically targets subcategorized arguments. Landau illustrates this proposed restriction to type e by examining constructions that feature an overt verb alongside silenced non-type e dependents such as nominal components of non-decomposable idioms, argumental adverbs, the name argument of naming verbs, and generalized quantifiers in argument positions — constructions that he claims are uniformly unacceptable. As Landau points out, there is no obvious reason why the semantic type of a subconstituent of a VP or other verbal projection should prevent the silencing of its container.

In this appendix, we will not discuss all the examples that Landau provides in support of his claims. We intend to review them more fully in a later work. We will also not address the question of whether AE might exist alongside VMV′E in Hebrew (and other languages with similar constructions), providing an alternative parse for some of the examples that we have analyzed as instances of VMV′E.[50]

We conclude with a novel conceptual point concerning the two proposals not previously considered in the debate between AE and V-stranding derivations. Suppose it should turn out that one or more of Landau's generalizations are

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50. We suspect that this is not the case; i.e., that AE does not coexist with VMV′E as a possible parse for object gaps. We believe it may be possible to analyze any genuinely acceptable multiple-object construction in which some, but not all, nominals are silenced as a construction whose overt arguments are overt simply because they have been raised out of V′, thereby escaping the effects of E-extension. An argument against such an analysis, as Idan Landau (personal communication) has reminded us, would be a construction with that character whose overt material can independently be shown to be unmovable (at least to the relevant position). We are not sure such cases exist, though Landau has suggested two candidates.

One is discussed by Landau (2020b, 352). He cites a contrast noted by Rouveret (2012, 928, ex. (69) and (70c)) between the unacceptability of an overt object-oriented depictive stranded by verbal ellipsis in English, with an auxiliary verb is the right survivor, seen in (i); and the acceptability of a Portuguese construction like (ii) in which the main verb and object-oriented depictive are overt but the direct object is not.

(i) *Lucy submitted the manuscript unfinished and Jan did badly typed.

(ii) O João devolveu o livro em bom estado e a Maria devolveu mas estragado
the John gave.back the book in good condition and the Maria gave.back more damaged
'João gave back the book in good condition and Maria gave it back more damaged.'

Landau, following the analysis favored by Rouveret, takes (i) to show that an object-oriented depictive is barred from raising out of the verb phrase in English and therefore probably also in Portuguese — making it impossible to analyze (ii) in the manner we would favor: as an instance of verb-stranding ellipsis in which the depictive remains overt thanks to raising outside the elided domain. On our approach to these constructions, however, the object-oriented depictive would have to move higher in (i) (above Aux′) than it would have to move in (ii) (above V′), which might be the crucial factor. (Something similar is noted by Rouveret himself as a disfavored, but possible alternative.)

Landau (2018, 19-20) also notes that the binding relations that may hold between a silent argument and an overt one in a ditransitive VP indicate that the overt argument escaped silencing via movement. The critical observation is the following: As shown in (iiia-c), the DP object of a ditransitive VP can bind a PP object that it precedes linearly (iiia), but not if it follows the PP linearly (iiib), despite the fact that scrambling of the PP to a position above DP (and linearized *before* DP) is otherwise available (iiic). In response to the question in (iiid-Q), however, one can felicitously utter (iiid-A) where, on our approach, the overt PP can only escape silencing via movement to a position above the head whose E-feature is extended to silence the DP. This would presumably predict iiib, where similar movement takes place, to be grammatical as well — contrary to fact.

(iii) a. hišveti et roš ha-memšala le-acmo (me-lifney šnatayim)
compared.1SG ACC head the-government to-himself (from-before two.years)
'I compared the prime minister to himself (from two years ago).'

b. * hišveti le-acmo et roš ha-memšala (me-lifney šnatayim)
compared.1SG to-himself ACC head the-government (from-before two.years)

c. hišveti la-nasi et roš ha-memšala
compared.1SG to.the-president ACC head the-government
'I compared the prime minister to the president.'

d. **Q:** le-mi hišveta et roš ha-memšala? **A:** hišveti le-acmo
to-who compared.2SG ACC head the-government compared.1SG to-himself
'Who did you compare the prime minister to? To himself.' Landau (2018, 19 ex. (34-35))

While we do not know why (iiib) is odd, we note that with a rich enough context and some help from a focus element, we can construct examples that are grammatical and have exactly the structure of (iiib). This is exemplified in (iv), which is felicitous, according to an author and our informants.

correct after all, and there are indeed semantic restrictions of some sort on the interpretation of the direct object in constructions that we have argued instantiate V-stranding V′ ellipsis. In the final section of this appendix, we suggest that a perspective on such problems developed by Pesetsky (2021) might provide a path forward, offering a means of reconciling such a finding with the analyses in this paper – by distinguishing between the related but distinct tasks of the speaker and hearer in the generation and interpretation of silenced material.

We begin with the empirical arguments.

Argument 1: the "Name" argument of naming verbs. One of Landau's arguments claimed to support AE over a head stranding ellipsis account of Hebrew object gaps rests on the observation that the "name" argument of a naming verb resists silencing when the verb remains overt in Hebrew, as illustrated in (89)-(90).

(89) **Name argument claimed to be unsilenceable**
*Hi kar'a la-xatul šela Geršon lifney še-ani karati la-xatul šeli __
She called to.the-cat her Gershon before that-I called to.the-cat my
Intended but unavailable: 'She called her cat Gershon before I called my cat Gershon.'

(90) **A:** Yosi kina et ha-ca'ad ha-ze ta'ut
Yosi dubbed ACC the-measure the-this mistake
'Yosi dubbed this measure a mistake.'

B: *Od anašim kinu et ha-ca'ad ha-ze __.
more people dubbed ACC the-measure the-this
Intended but unavailable: 'More people did so.' (Landau 2022, 815, ex. (53)-(54))

Crucially, the name argument in examples like these has been shown by Matushansky (2008) to exhibit syntactic hallmarks of predicative arguments, and thus to be of a functional semantic type. On Landau's approach to Hebrew object gaps, AE is the only mechanism available for silencing verbal objects when the verb is itself overt, and this mechanism is restricted to objects of non-functional types. Thus, the name arguments of naming verbs are *expected* to resist silencing via AE.

However, it seems to us that examples like those in (89)-(90) do not have the right properties to test Landau's claim, since the attempted silencing operation in these examples targets *only* the name argument of the ditransitive verbs 'call' and 'dub', to the exclusion of their second internal argument, which denotes the entity being named. Examples like (89)-(90) do not have a parse involving verb stranding and verb-projection ellipsis, which necessarily involves the silencing of all verbal arguments – except, of course, if one of these arguments has itself vacated the silenced projection and therefore remains unsilenced.

In fact, it turns out that such examples become fully grammatical once *both* of the verbal objects are silenced rather than just the name argument, e.g., (91)-(92). In the context of, say, an argument over who first came up with the name *Gershon* as a way to call my cat, (91) is a perfectly acceptable contribution to the discourse. Similarly, (92) is an acceptable discourse in the context of figuring out who was the first to warn that taking the measure discussed would be a mistake.

(iv) efšar lehašvot rašey memšala kodmim le-harbe dmuyot historiyot, aval ata yaxol lehašvot rak
possible to.compare head.PL government previous to-many characters historical, but you can compare only
le-acmo et roš ha-memšala ha-noxexi
to-himself ACC head the-government the-present
'One can compare previous prime ministers to many historical figures, but you can compare only to himself the current prime minister.'

This suggests to us that movement of the reflexive-embedding PP should not *in principle* disrupt the binding relations between the PP and DP arguments, and that there is an alternative source for the infelicity of (iiib). One conjecture might be that focus is a precondition for the raising of the PP, and that in (iiib) in particular, there is no sufficient help from focus to license it, in contrast to (iv)

We leave the necessary fuller exploration of this issue for future work. We merely mention it as an important question that our defense of V-stranding ellipsis raises about the full repertoire of silencing operations available to the grammar. See also the final subsection of this appendix for other relevant discussion.

(91) **Name argument silenceable if other argument is also silenced**
Ani karati la-xatul šeli Geršon lifney še-at karat __.
I called to.the-cat my Gershon before that-you called
'I called my cat Gershon before you did.'

(92) **A:** Yosi kina et ha-ca'ad ha-ze ta'ut
Yosi dubbed ACC the-measure the-this mistake
'Yosi dubbed this measure a mistake.'

B: Od anašim kinu __ lefanav.
more people dubbed before.him
'More people did before him.'

Furthermore, and as predicted by our approach, the name argument may be the only one silenced if we ensure that the other internal argument has vacated the verbal projection targeted for deletion. This is the case when that argument has undergone *topicalization* as in (93), or has raised to satisfy parallelism constraints in constructions involving antecedent-contained deletion as in (94).[51]

(93) **Name argument silenceable if other argument has moved out of silenced domain**
la-madrix ha-macxik šeli ba-cofim karati štutnik, aval la-marce ha-kašu'ax šeli
to.the-counselor the-funny mine in.the-scouts called.1SG goofball, but to.the-lecturer the-strict mine
ba-universita lo karati __.
in.the-university NEG called.1SG
'My funny scout counselor, I called a goofball, but my strict university lecturer I did not.'

(94) ani kiniti ta'ut kol ca'ad še-ata kinita __.
I dubbed mistake every measure that-you dubbed
'I dubbed a mistake every measure that you did <dub a mistake>.'

This fuller paradigm actually posits a forceful challenge to Landau's theory: If AE is possible and is the only mechanism available to us for silencing objects under an overt lexical verb, then (91)-(92) must be derived by applying AE separately to the two internal arguments of the naming verbs in these examples, and (93)-(94) must be derived by applying AE to the name argument in particular. But as Landau illustrates with (89)-(90) – this should be ruled out by the ban on AE of functional arguments. On the other hand, the totality of the data is predicted by the approach we have advanced here. Crucially, the infelicitous cases of silencing (89)-(90) are exactly those cases that cannot be derived via VMV′E for independent reasons, and therefore lack any grammatical parses.

Argument 2: Generalized quantifiers – the case of $\forall \gg \exists$. Landau argues that generalized quantifiers (GQs) can undergo silencing under an overt lexical verb only if the resulting object gap can be construed as containing not a GQ, but rather a pronominal element of type e. For example, while (95) does appear to involve the silencing of a generalized universal quantifier, this is possible, on Landau's theory, because the result is equivalent to a sentence in which instead of an object gap, a pronoun is used — and that pronoun denotes the individuals that the overt GQ in the ellipsis antecedent successfully individuates. Crucially, both the pronoun and the gap in (95) denote the plural individual containing every individual in the domain of the anteceding universal quantifier, and lack any quantificational force of their own.

(95) **Silenced element interpretable as pronoun (*per* Landau 2023)) …**
Ani makir kol student ba-kita ha-zot. Gam ata makir ___ .
Ani makir kol student ba-kita ha-zot. Gam ata makir **otam**.
I know every student in.the-class the-this. Also you know them
'I know every student in this class. You also know them.'

This approach immediately predicts that an object gap ostensibly corresponding to a universal quantifier will be extremely limited in its scopal interaction with other scope-taking elements in the sentence. In particular, a silenced

51. We thank Danny Fox for this observation.

universal quantifier under an overt lexical verb should not take scope and distribute over an existential quantifier higher in the structure. This prediction seems to be borne out, as illustrated in (96), where an overt universal quantifier gives rise to an inverse scope reading relative to an existential, while an elided one reportedly does not. As further predicted by Landau's type restriction on silenced arguments, the object gap below patterns like an overt pronoun, which, as shown in (96b), also does not give rise to distributive readings.

(96) a. **…correctly predicting no inverse scope (claim by Landau 2023, 14)**

Ba-xodeš	še-avar,	lakoax	(šone)	hizmin	kol	parit	ba-xanut.
in.the-month	that-passed	costumer	different	ordered	every	item	in.the-store

'Last month, a different costumer ordered every item in the store.' $\forall \gg \exists, \exists \gg \forall$

b.

Gam	ba-xodeš	ha-ze,	lakoax	(?šone)	hizmin	__ / otam.
also	in.the-month	the-this	costumer	different	ordered	/ them

'This month too, a *specific* (different) costumer ordered every item.' $*\forall \gg \exists, \exists \gg \forall$

Nonetheless, here too we believe that a fuller view of the data actually provides an argument *against* Landau's approach, similar in structure to the argument from naming predicates just discussed above. First, we show that the absence of a wide, inverse-scope reading of (96b) does not constitute an argument for adopting AE as the only mechanism that can derive object gaps in Hebrew. We do so by illustrating that the inverse-scope reading of (96b) would be ruled out on both an AE and and VMV′E parse, for different reasons in each case. We then show that once the example is modified in a way that makes a VMV′E parse available for the inverse-scope reading, the result is fully acceptable and the silenced object gap behaves as if it were an overt universal quantifier, not a pronoun.

Landau argues that the absence of an inverse-scope reading of (96b) remains unaccounted for under any verb-stranding ellipsis account, since generalized quantifiers embedded in elided constituents do not generally lose their ability to take scope. We contest this conclusion as well as the empirical generalization used to justify it. We agree with Landau that silencing a constituent does not categorically restrict the scope possibilities of its scope-taking subconstituents. This is not the end of the story, however. The inverse-scope reading that (96) lacks is equally absent when we attempt to reformulate the example as a standard example of (what previous work would have called) VP-ellipsis in which the right-survivor is crucially an auxiliary verb, not a main verb. Such an example cannot be parsed as an instance of AE, since the main verb has been silenced along with its complements. We demonstrate this new observation below by embedding a silenced verbal projection whose object is a universal quantifier under the Hebrew auxiliary *haya* in both its uses (see section 7 above): habitual in (97) and counterfactual (X-marking) in (98).

(97) **Inverse scope also impossible in verb-projection ellipsis where AE analysis cannot be maintained**

a.

Be-šnot	ha-tiš'im,	mefake'ax	(šone)	haya	mevaker	be-hafta'a	kol	xanut	šel
in.the-years	the-ninety,	inspector	different	*haya*.HAB	visit	in-surprise	every	store	of
ha-rešet	hazot.								
the-chain	this								

'In the nineties, a different inspector used to visit every store in this chain unannounced.'
$\forall \gg \exists, \exists \gg \forall$

b.

Gam	be-šnot	ha-alpayim,	mefake'ax	(?šone)	haya	__.
also	in.the-years	the-two.thousand,	inspector	different	*haya*.HAB	

'In the two thousands, a specific (different) inspector also used to.' $*\forall \gg \exists, \exists \gg \forall$

(98) a. ilu haya lanu khal lakoxot gadol yoter, lakoax (šone) haya mazmin kol
had *haya*.X to.us audience costumers big more, costumer different *haya*.X order every
parit ba-xanut.
item in.the-store
'If we had a larger costumer base, a costumer would have ordered every item in the store.'
$\forall \gg \exists, \exists \gg \forall$

b. Gam ilu hayinu moxrim paxot pritim, uli lakoax (?šone) haya __
also if *haya*.X.1PL sell less items maybe costumer different *haya*.X
'Had we offered less items, maybe a *specific* (different) costumer also would have.' $*\forall \gg \exists, \exists \gg \forall$

If the factor that blocks inverse-scope in (96) also blocks it in (97)-(98) (the optimal hypothesis), it cannot be a constraint specifically targeting object gaps selected by an overt lexical verb. As it happens, there is an independent constraint on the size of elided constituents in certain configurations that covers the effects in all the examples of (96)-(98): MAXELIDE (Merchant 2008 [circulated 2001]; Takahashi & Fox 2005), a simplified version of which is stated in (99). The missing inverse-scope reading in each case would require a derivation that violates this constraint;[52] This constraint was originally intended to account for the unacceptability of VP-ellipsis when sluicing is available, illustrated in (100).[53] It has also been invoked, however, to explain the interaction of quantifier scope and ellipsis, given that scope taking via quantifier raising (QR) creates chains of binders and bindees throughout the structure that can meaningfully interact with MAXELIDE, as discussed by Takahashi & Fox (2005).

(99) **MAXELIDE**
Ellipsis of a constituent XP is disallowed if:
1. XP contains a variable x whose binder λ_x is external to XP, and
2. the smallest constituent YP that dominates XP and contains both x and λ_x is
   (a) semantically identical to some anteceding constituent, modulo focus-marked constituents; and
   (b) deletable or contains a deletable constituent that dominates XP.

(100) Bill read several books on this list, but I don't know which books (*he did) __.

In light of this constraint, let us examine Landau's examples (96a-b) again. On the inverse-scope construal of (96b), the universal quantifier undergoes QR to a position above the indefinite subject, and the antecedent and ellipsis clauses have the structures schematized in (101), where QR creates a binder-bindee relation between a GQ and a trace-converted version of its copy (Fox 1999; 2002).[54]

(101) a. **Antecedent:** [Last month [every item [λx [a costumer [ordered [λy [y [the$_x$ item]]]]]]]]
b. **Ellipsis:** [And this month also [every item [$_{\text{EC}_1}$ λz [a costumer [ordered [$_{\text{EC}_2}$ λr [r [the$_z$ item]]]]]]]]

To derive the unacceptable sentence in (96b) on a VMV′E account, what must be elided is the constituent labeled EC_2 ("Elided Constituent") in (101b). However, EC_2 contains a variable whose binder is external to it, and the smallest constituent containing the binder and its bindee, labled EC_1 in (101b), is itself deletable in Hebrew by means of *stripping* (perhaps a variant of sluicing, with E-extension from a null C), as illustrated by the felicity of the discourse in (102). Consequently, the unacceptability of Landau's (96) is unproblematic under a VMV′E approach. Selecting EC_2 rather than EC_1 for silencing is ruled out independently by MAXELIDE. This explanation extends, without any further assumptions, to the absence of an inverse-scope reading of (97)-(98).

52. We follow the version of MAXELIDE advanced in Takahashi & Fox (2005), a development of Merchant's (2008 [circulated 2001]) earlier proposal.

53. In (100) the smallest constituent containing both the wh-phrase and its trace – namely, the binder and the bindee – contains a TP that is deletable via sluicing, ruling out a smaller ellipsis operation targeting only the VP

54. In (101) we assume that head movement too creates a binder-bindee relation between the raised verb and its trace. This has been argued to yield MAXELIDE effects by Hartman (2011), though this is not a crucial assumption for our purposes.

(102) **Stripping (high ellipsis) permits inverse scope**

a. Ba-xodeš še-avar, lakoax (šone) hizmin kol parit ba-xanut.
in.the-month that-passed costumer different ordered every item in.the-store
'Last month, a different costumer ordered every item in the store.' $\forall \gg \exists, \exists \gg \forall$

b. (Le-ma'ase,) gam ba-xodeš ha-ze __.
in-fact also in.the-month the-this
'In fact, this happened this month too.' $\forall \gg \exists, \exists \gg \forall$

That it is in fact MaxElide which rules out inverse-scope in these examples is supported by the fact that when ellipsis of a verbal projection does not violate MaxElide, the inverse-scope reading is acceptable even for universal quantifiers that are silenced while the governing verb remains overt. In particular, Takahashi and Fox (2005) illustrate with examples (103) that intervening focus between a binder and its variable obviates MaxElide effects. They attribute this fact to a general ban on silencing focused material, with the consequence that an intervening focused constituent prevents ellipsis of any constituent dominating it. Focused negation in an example like (103) thus rules out sluicing, rendering VP the largest deletable constituent in (103) — and thus rendering its deletion compliant with MaxElide.

(103) **Focused elements count as undeletable for MaxElide and block its effect**
I cannot tell you which books Bill read, but I can certainly tell you which books he did NOT __.

Extending this observation to our domain, we make the following prediction: in a sentence with an indefinite subject and a universal quantifier in object position, placing contrastive focus on the indefinite subject would result in a structure in which silencing a verbal projection would not be ruled out by MaxElide. This should make a wide-scope construal of the universal quantifier acceptable. This is confirmed in (104), which seems to us fully acceptable on an inverse-scope reading of the object gap. We again schematize the structure of the elided constituent and its anteceding material as in (105). Crucially, because of the focused indefinite in (105b), EC_1 is ruled out as a potential ellipsis site. Consequently eliding EC_2 does not result in a MaxElide violation, and an inverse scope reading of the universal is achieved even though VMV′E has taken place.

(104) **Where MaxElide effects are blocked by focus, inverse scope possibiliy reappears**
AXOT (šona) badka kol xole ba-xeder miyun ve rak az gam ROFE (šone)
NURSE (different) checked every patient in.the-room sorting and only then also DOCTOR (different)
sof-sof badak __.
finally checked
'A nurse checked every patient in the ER and only than a Doctor did.'

(105) a. **Antecedent:** [every patient [λx [a NURSE [checked [λy [y [the$_x$ patient]]]]]]]

b. **Ellipsis:** [every patient [$_{\mathrm{EC}_1}$ λz [a DOCTOR [checked [$_{\mathrm{EC}_2}$ λr [r [the$_z$ patient]]]]]]]

This result not only shows that the scope possibilities in (96) are explained under a VMV′E account of object gaps, but also posits a direct challenge to Landau's account that derives object gaps *only* via a type-restricted operation — mistakenly excluding wide-scope for an object universal even if the subject above it is focused.

But what if…? Exploring the hypothetical. In the previous subsection, we examined two of Landau's arguments in favor of an AE-only analysis of Hebrew object gaps, and argued that the phenomena under discussion, when examined in greater detail, actually support the verb-stranding ellipsis proposal advanced here. As noted above, we leave for later work a fuller exploration of Landau's many other arguments.[55] These arguments are relevant to our proposals because, as we also noted above, we do not expect a constraint on the process of V′-ellipsis to care about the semantics of a nominal subconstituent.

55. The possibility that Hebrew has an independent null object strategy that is not parasitic on verb movement is one that has also been advanced by advocates of verb-stranding VP ellipsis approaches such as Doron (1999, 127 ff.;1990) and Goldberg (2005).

We do believe that the fuller picture does not uniformly support his claim that verb-stranding ellipsis proposals overgenerate, while AE with its semantic restrictions does not.[56] But imagine nonetheless, for the sake of the argument, that our skepticism turns out to be wrong in whole or in part, and suppose there truly are semantic restrictions that specifically concern silent object nominals in constructions like those under discussion. In this final section, we would like to suggest that such a discovery would still not necessarily militate against our claim that such structures instantiate VMV′E rather than AE.

They key to our proposal is the observation that every silent constituent in an utterance imposes a burden on the hearer of that utterance, who must supply an identity for it. This is true regardless of the grammatical source of the silencing: another sense in which we observe a ”uniformity of silence”. If the entirety of a V′ is silent due to ellipsis, the hearer must produce a working hypothesis concerning the identity of both the silent verb and its silent complement. But if the verb is overt and only the complement is silent, as in the constructions discussed in this appendix, the hearer's task is limited to that complement. This is the very fact that makes it possible to entertain multiple theories in the first place about what has been elided, like the two debated in this appendix. We would like to suggest that this observation could be the source of any limitations on the nature of the silenced complement that might turn out to be real, as we elaborate below.

We take it for granted that when a speaker generates a structure containing E-bearing silent constituents, it should comport with the rules of grammar that generate such structures. It is these rules that have constituted the topic of this paper so far. The hearer of an utterance resulting from the generation of such a structure, in turn, has the familiar and much-discussed task of "reverse engineering" the speaker's derivation — consistent with the same rules of grammar, of course. The existence of the E-feature and the rule of E-extension, however, with their power to silence entire constituents, places a special burden on the reverse engineering process. The hearer must not merely reverse engineer the structure recursively built from the lexical items that they have perceived in the speaker's utterance, but must also form a working hypothesis concerning the identity of the lexical items whose identity has been masked by silencing.

This observation immediately raises the question: does the human language faculty impose constraints on the reverse engineering of silenced material under such circumstances? We already took note of this question in the final paragraph of the introductory section of this paper, where we hinted at the possibility that much of the literature devoted to questions of "parallelism requirements, interactions with scope and anaphoric relations, and much more" might be understood as concerning itself less with the generation of ellipsis structures and more with the laws that govern the reverse engineering of such structures by a hearer confronted with silenced material.

We returned to that question briefly in our discussion of voice mismatches, in connection with the structure in (74) and our hypothesis that the locus of voice distinctions in the English passive is the copular auxiliary be_{PASS}.

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56. A few brief responses to some of these arguments follow:

a. Landau (2022, 805ff) notes that an object participating in a non-decomposable idiom may not be silenced while the verb remains overt, attributing this to the failure of such a nominal to have the semantic type e. We might attribute this to a uniformity requirement on the pieces of an idiom that requires either all lexical items in the idiom to be overt (in some structural position) or none of them. Additionally, some of Landau's examples once again involve ditransitive verbs, and once again, silencing both objects rather than just one significantly improves the example in question.

b. Landau (p.811 ff) shows that Hebrew obligatory adverbs (counterparts to English *treat someone* *(*well*) etc.) resist silencing when the verb remains overt. Viewing such adverbs as argumental, Landau assumes that they are merged as sisters of the verb, and that the impossibility of selectively silencing them once again stems from their semantic status as not type e. Alexiadou (1997) has argued that such adverbs — manner and locative adverbs more generally, in fact — do indeed merge first as complements, but obligatorily raise into a higher domain (by a kind of A-movement similar in nature and motivation to object shift for nominal objects). If some version of this proposal is correct (or alternatively, if obligatory adverbs are merged outside V′, their obligatoriness following from some other factor), then Landau's observation is just another instance of adjunct exclusion, for which we have already provided an account. Note that the same adverb may simultaneously modify a verb that requires it and a verb that does not in a conjunction structure: e.g. *They fed and treated me well*, *She spoke and behaved badly*), suggesting a common syntax.

   A similar account might be advanced for measure-phrase arguments of verbs like 'weigh', though at present we find this claim more difficult to justify independently.

c. Landau (p.816) claims that Hebrew predicate nominals in expressions like 'she became a manager' may not be selectively silenced in the presence of an overt verb. The first author and our consultants disagree with this claim.

There we considered the implications for the hearer when E-extension applies from the trace of a raised copular auxiliary, silencing the minimal phrase that contains it. As we noted, the lexical identity of the raised verb is in no doubt in such a configuration, and poses no special problem for the hearer, since it remains fully audible in its new position outside the elided domain. It is only the remainder of the elided material that poses a special problem for the hearer, who must form a hypothesis concerning the identity of the lexical items that form the subcategorized complement of the raised verb. For this reason, we proposed that it is only the complement of the raised passive auxiliary that must be syntactically identical to a linguistic antecedent — deriving in this way the possibility of an active-passive mismatch between the ellipsis site and the antecedent (given our discussion of participial morphology as goal flagging).

Now consider the hearer's task when a *main* verb raises and E-extension applies from its trace (our analysis of the object-gap constructions that Landau argues instantiate AE). The logic of our discussion of passive-auxiliary raising holds here as well. The hearer is *not* faced with the task of reverse-engineering the identity of the raised main verb, since it remains fully audible in its new position. The only task that faces the hearer in this connection concerns the complement of the main verb and what material the speaker might have used to generate that complement. Putting the matter differently, though the speaker generated verb-projection ellipsis, the problem that the hearer has to solve amounts to argument ellipsis.

Thus, if it should turn out that there are hearer-side restrictions particular to the task of figuring out the nature of a silenced argument in a situation where the predicate's identity is not at issue, our analysis could be reconciled with observations like those advanced by Landau. Our proposals could be the correct description of the speaker-side generation of ellipsis structures, while Landau's observations would be taken to concern the nature of what may (and may not) be posited by a hearer faced with an overt verb and a silent complement.[57]

A question now arises concerning situations in which both a verb and its complement are fully silenced, as happens in sluicing constructions, or when an auxiliary verb is the right-survivor in a verbal ellipsis construction. In such situations, no claims have been made in the literature that the complement of a fully silenced verb is restricted to type-*e* semantics (or anything similar). The various phenomena discussed in this connection by Landau all involve overt verbs coupled with silent objects — never situations in which both are silent. The hearer must therefore have the ability to reverse-engineer the derivation of a fully silent verbal projection without any special restrictions on the verb's object. This raises a question (continuing to accept the hypothetical that such restrictions exist): why cannot this same ability be put to work even when the verb has raised out of the silenced domain, leaving it overt?

A possible answer might lie in a general property of hearer's strategies in dealing with silent material explored by Pesetsky (2021), who argued that the human language faculty limits a hearer to maximally "unambitious" reconstructions of silent material, in accordance with a principle he called the Principle of Unambitious Reverse Engineering (PURE):

(106) **Principle of Unambitious Reverse Engineering (PURE)**
When determining the identity of unpronounced material in the course of reverse-engineering a speaker's syntactic derivation, the language system of the hearer considers only the minimally semantically contentful possibilities compatible with the morphosyntactic environment. (Pesetsky 2021, 196)

PURE was argued to be responsible for limitations on the tense interpretation of silenced fronted auxiliaries in colloquial English yes/no questions (*You own that car?*) (the so-called factative effect discussed by Fitzpatrck 2006), as well as the tense and aspect properties of English nonfinite clauses (reanalyzing discoveries by Wurmbrand 2014). Should it turn out that some version of Landau's generalizations concerning silent objects is sustainable, they might similarly be attributable to PURE. Additionally, however, the hearer's inability to reverse-engineer a partially silenced V′ with the same freedom available to the reverse-engineering of a fully silenced V′ might be understandable as another example of "unambitiousness", in the spirit (if not the letter) of PURE as formulated above — requiring the reverse-engineering process to limit itself to the minimal syntactic domain within which it is necessary.

57. The term "hearer" should be understood in the same fashion and with the same caveats discussed by Pesetsky (2021, 196), who cautions: "I will [...] refer directly to the hearer's language system with the word "hearer" to keep the prose simple — but it is the hearer's language system that I intend throughout. I also assume that a speaker's language system self-monitors in the process of speech production, functioning as hearer as well as speaker, so that the planning of an utterance takes into account the restricted range of interpretations a hearer is permitted to entertain by the principle proposed below. So the term *hearer* in this paper stands for an emphatically abstract concept."

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